ROYAL STATISTICAL SOCIETY
DATA | EVIDENCE | DECISIONS

Journal of the Royal Statistical Society
Statistical Methodology
Series B

J. R. Statist. Soc. B (2015)
77, *Part* 5, *pp.* 947–972

A split-and-merge Bayesian variable selection approach for ultrahigh dimensional regression

Qifan Song

Purdue University, West Lafayette, USA

and Faming Liang

University of Florida, Gainesville, USA

[Received May 2013. Final revision August 2014]

Summary. We propose a Bayesian variable selection approach for ultrahigh dimensional linear regression based on the strategy of split and merge. The approach proposed consists of two stages: split the ultrahigh dimensional data set into a number of lower dimensional subsets and select relevant variables from each of the subsets, and aggregate the variables selected from each subset and then select relevant variables from the aggregated data set. Since the approach proposed has an embarrassingly parallel structure, it can be easily implemented in a parallel architecture and applied to big data problems with millions or more of explanatory variables. Under mild conditions, we show that the approach proposed is consistent, i.e. the true explanatory variables can be correctly identified by the approach as the sample size becomes large. Extensive comparisons of the approach proposed have been made with penalized likelihood approaches, such as the lasso, elastic net, sure independence screening and iterative sure independence screening. The numerical results show that the approach proposed generally outperforms penalized likelihood approaches: the models selected by the approach tend to be more sparse and closer to the true model.

Keywords: Bayesian variable selection; Posterior consistency; Split and merge; Stochastic approximation Monte Carlo sampling

1. Introduction

Consider the problem of variable selection for the linear regression model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}, \tag{1}$$

where $\mathbf{y} = (y_1, \ldots, y_n)^{\mathrm{T}} \in \mathbb{R}^n$ is the response vector, n is the sample size, $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_{P_n}) \in \mathbb{R}^{n \times P_n}$ is the design matrix, P_n is the number of explanatory variables (which are also known as predictors), $\boldsymbol{\beta}^*$ is the vector of unknown regression coefficients and $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^{\mathrm{T}} \sim N_n(0, \sigma^{*2})$ is Gaussian random error. In this paper, we use the bold letter $\mathbf{x}_i$ to denote an n-vector realization of the ith predictor. The subscript n in P_n means that P_n is a non-decreasing function of n. When P_n is large, identification of the true predictors for model (1) is of particular interest, as this can generally improve the accuracy of prediction for future observations and deepen one's understanding of the underlying system. In this paper, our study of variable selection will focus on the scenario $P_n \gg n$, i.e. the column dimension of $\mathbf{X}$ is ultrahigh. This kind of problem has

Address for correspondence: F. Liang, Department of Biostatistics, University of Florida, Gainesville, FL 32610, USA.
E-mail: faliang@ufl.edu

© 2014 Royal Statistical Society 1369–7412/15/77947

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

arisen in diverse fields, ranging from biomedical science to social science. For instance, in a genomewide association study one may wish to identify the genes or single nucleotide polymorphisms (SNPs) that are associated with a certain disease, and in economic study one may wish to identify stocks that generate profits in investment portfolios.

In the literature, the problem of variable selection for high dimensional regression is often treated with penalized likelihood approaches, which are to find a model ξ that minimizes the penalized likelihood function

$$-\log\{f(\mathbf{y};\mathbf{X}_\xi,\boldsymbol{\beta}_\xi)\}+p_\lambda(\boldsymbol{\beta}_\xi),$$

where $\mathbf{X}_\xi$ and $\boldsymbol{\beta}_\xi$ denote respectively the design matrix and the regression coefficient vector corresponding to the model ξ, and $p_\lambda(\cdot)$ is a penalty function with tuning parameter λ which can be determined through cross-validation. The popular penalized likelihood approaches include the lasso (Tibshirani, 1996), the adaptive lasso (Zou, 2006), the elastic net (Zou and Hastie, 2005), smoothly clipped absolute deviation (SCAD) (Fan and Li, 2001) and the minimax concave penalty (Zhang, 2010), among others. The lasso and adaptive lasso employ an L_1-penalty function, the elastic net employs a linear combination of an L_1-penalty and an L_2-penalty, and SCAD and the minimax concave penalty employ some concave penalty functions.

Although these penalized likelihood approaches have been successfully applied to many problems, their performance tends to deteriorate as P_n increases. To address this issue, Fan and Lv (2008) and Fan and Song (2010) proposed the sure independence screening (SIS) approach and an iterative version of SIS (ISIS). SIS first reduces the dimension of the problem by screening out those predictors with low marginal utility and then refines the variable selection by using the lasso or SCAD. The marginal utility measures the usefulness of a single predictor for predicting the response. For linear regression, it is simply measured by the correlation between each predictor and the response. To avoid missing true predictors, ISIS iteratively selects predictors from the remaining unselected predictors. Compared with other penalized likelihood approaches, SIS and ISIS are more efficient in computation as they work on a dimension-reduced space. Moreover, SIS and ISIS tend to outperform other penalized likelihood approaches in variable selection; the models selected by SIS and ISIS tend to be more sparse.

Other than the penalty functions that were mentioned above, Chen and Chen (2008) proposed the extended Bayesian information criterion (EBIC) with an L_0-penalty function given by

$$p_\lambda(\boldsymbol{\beta}_\xi)=\frac{|\xi|}{2}\log(n)+\lambda|\xi|\log(P_n),$$

where $|\xi|$ denotes the number of predictors contained in the model ξ. Chen and Chen (2008) showed that, if $P_n=O(n^\kappa)$ and $\lambda>1-1/(2\kappa)$, then EBIC is consistent, i.e. the minimum EBIC model will converge to the true model in probability as the sample size becomes large. A major concern with the EBIC is the lack of an efficient optimization algorithm to search over the model space. Because the model space is discrete, computer-intensive algorithms, such as Markov chain Monte Carlo (MCMC) algorithms, must be used.

Parallel to penalized likelihood approaches, various Bayesian approaches have been developed to tackle this problem. These approaches advance either in a sophisticated model search algorithm, such as evolutionary stochastic search (Bottolo and Richardson, 2010), or in a prior specification that is particularly suitable for high dimensional problems, such as the lasso prior (Park and Casella, 2008), the horseshoe prior (Carvalho *et al.*, 2010; Polson and Scott, 2012), the Ising prior (Li and Zhang, 2010), the proportion of variance explanation prior (Guan and Stephens, 2011) and the non-local prior (Johnson and Rossel, 2012; Johnson, 2012). Although various numerical examples have demonstrated the successes of Bayesian approaches, there are

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

no extensive studies to show that Bayesian approaches can lead to a consistent selection of true predictors in $P_n > n$ problems until quite recently; see for example Kundu and Dunson (2014) and Bondell and Reich (2012). Strawn *et al*. (2012) also studied the consistency of regression coefficient estimation in $P_n > n$ problems. A major concern with Bayesian approaches is their high computational demand. Since the volume of the model space increases geometrically with the dimension P_n, the central processor unit (CPU) time for a Bayesian approach should increase accordingly or even faster.

In this paper, we propose Bayesian variable selection based on the strategy of the split-and-merge (SAM) approach. The approach proposed consists of two stages.

(a) Split the high dimensional data into a number of lower dimensional subsets and perform Bayesian variable selection for each subset.
(b) Aggregate the variables that are selected from each subset and perform Bayesian variable selection for the aggregated data set.

Under mild conditions, we show that the approach proposed is consistent, i.e. the true model will be identified in probability 1 as the sample size becomes large, and that the approach proposed has favourable time complexity compared with conventional Bayesian approaches and even penalized likelihood approaches. Henceforth, we shall call the approach proposed the SAM approach for simplicity of description.

Our contribution in this paper is twofold. First, we propose a computationally feasible Bayesian approach for ultrahigh dimensional regression. Since the SAM approach has an embarrassingly parallel structure, i.e. it can break the task of high dimensional variable selection into many small tasks between which there is a dependence or communication, it can be easily implemented in a parallel architecture and applied to big data problems with millions or more of predictors. This has been beyond the ability of conventional Bayesian approaches, as they directly work on the full data set. Second, under mild conditions we show that the Bayesian approach proposed can share the same asymptotics, such as sure screening and model selection consistency, as SIS. In spirit, the SAM approach is similar to SIS and ISIS. Its first stage serves the purpose of dimension reduction, which screens out uncorrelated predictors; and its second stage refines the selection of predictors. However, compared with SIS and ISIS, the SAM approach can often lead to more accurate selection of true predictors. This is because it screens uncorrelated predictors on the basis of the marginal inclusion probability, which has incorporated the joint information of all predictors contained in a subset. Whereas, as previously mentioned, SIS makes use of only the marginal information of each predictor. ISIS tries to incorporate information from other predictors, but in an indirect way.

Finally, we note that the SAM strategy, or otherwise known as divide and conquer, has been often used in big data analysis; see for example Mackey *et al*. (2012), Lin and Xi (2011) and Xi *et al*. (2009). Our use of this strategy is different from others. In the SAM approach, the data split is done in the dimension of predictors, whereas this is done in the dimension of observations in other work. This difference necessitates the development of new theory for variable selection under the situation that the model is misspecified because of missing some true predictors. This will be detailed in Section 2.3.

The rest of this paper is organized as follows. Section 2 establishes consistency of the SAM approach for variable selection. Section 3 summarizes the approach and discusses some implementational issues. Section 4 illustrates its performance by using two simulated examples along with comparisons with penalized likelihood approaches. Section 5 compares the SAM approach with penalized likelihood approaches on two real data examples. Section 6 concludes the paper with a brief discussion.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

The data that are analysed in the paper and the programs that were used to analyse them can be obtained from

http://wileyonlinelibrary.com/journal/rss-datasets

2. Variable selection consistency of the split-and-merge approach

This section is organized as follows. In Section 2.1, we establish the posterior consistency for correctly specified models, i.e., if the model is correctly specified with all true predictors included in the set of candidate predictors, then the true density of observations from model (1) can be consistently estimated on the basis of the models sampled from the posterior as the sample size becomes large. The theory development and conditions in this subsection follow Jiang (2007) very closely. In Section 2.2, we establish the sure screening property for the correctly specified models on the basis of the marginal inclusion probability, i.e. the marginal inclusion probability of the true predictors will converge to 1 in probability as the sample size becomes large. We also establish the consistency of the maximum *a posteriori* (MAP) model. In Section 2.3, we show that, for the misspecified models for which some true predictors are missed in the set of candidate predictors, the sure screening property based on the marginal inclusion probability still holds. Combining the theory that is established in Sections 2.1–2.3 leads to the variable selection consistency of the SAM approach.

2.1. Posterior consistency for correctly specified models

Let $D_n = \{\mathbf{y}; \mathbf{x}_1, \ldots, \mathbf{x}_{P_n}\}$ denote a data set of n observations drawn from model (1). Let $x = (x_1, \ldots, x_{P_n})^{\mathrm{T}}$ denote a generic observation of the P_n predictors, and let y denote a generic observation of the response variable corresponding to x. Let ν_x denote the probability measure of x. Then the true probability density of (y, x) is given by

$$f^*(y,x)\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{\sqrt{(2\pi)}\sigma^*}\exp\left\{-\frac{(y-x^{\mathrm{T}}\boldsymbol{\beta}^*)^2}{2\sigma^{*2}}\right\}\nu_x(\mathrm{d}x)\,\mathrm{d}y = \phi(y; x^{\mathrm{T}}\boldsymbol{\beta}^*, \sigma^*)\,\nu_x(\mathrm{d}x)\,\mathrm{d}y, \tag{2}$$

where $\phi(\cdot; \mu, \sigma)$ denotes a Gaussian density function with mean μ and standard deviation σ.

As in conventional studies of high dimensional variable selection, we consider the asymptotics of the SAM approach under the assumption that P_n increases as n increases. To be specific, we assume that $P_n \succ n^\theta$ for some $\theta > 0$, where $b_n \succ a_n$ means $\lim_{n\to\infty}(a_n/b_n) = 0$. In addition, we assume that the true model is sparse. Two sparseness conditions are considered in this paper. One states that most components of $\boldsymbol{\beta}^*$ are very small such that

$$\lim_{n\to\infty} \sum_{j=1}^{P_n} |\beta_j^*| < \infty, \tag{3}$$

where β_j^* denotes the jth entry of $\boldsymbol{\beta}^*$. The other one is stronger but more usual, which requires that most components of $\boldsymbol{\beta}^*$ are 0 such that

$$\lim_{n\to\infty} \sum_{j=1}^{P_n} \mathbf{1}\{\beta_j^* \neq 0\} < \infty. \tag{4}$$

Let $\xi \subset \{1, \ldots, P_n\}$ denote a subset model, which includes the indices of all selected predictors. Let $|\xi|$ denote the number of predictors included in ξ. Thus, the strict sparsity condition (4) implies that the true model $\mathbf{t} = \{i : \beta_i^* \neq 0, 1 \leqslant i \leqslant P_n\}$ has a finite model size. For a model $\xi = \{i_1, i_2, \ldots, i_{|\xi|}\}$, we let $\mathbf{X}_\xi = (\mathbf{x}_{i_1}, \mathbf{x}_{i_2}, \ldots, \mathbf{x}_{i_{|\xi|}})$ denote the design matrix, and let $\boldsymbol{\beta}_\xi = (\beta_{i_1}, \ldots, \beta_{i_{|\xi|}})^{\mathrm{T}}$ denote the regression coefficient vector. Let $L(D_n|\xi, \boldsymbol{\beta}_\xi, \sigma^2)$ denote the

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

likelihood function, and let $\pi(\xi, \mathrm{d}\boldsymbol{\beta}_\xi, \mathrm{d}\sigma^2)$ denote the joint prior distribution imposed on ξ and the parameters σ^2 and $\boldsymbol{\beta}_\xi$. Then the posterior distribution is given by

$$\pi(\xi, \mathrm{d}\boldsymbol{\beta}_\xi, \mathrm{d}\sigma^2 | D_n) = L(D_n|\xi, \boldsymbol{\beta}_\xi, \sigma^2)\, \pi(\xi, \mathrm{d}\boldsymbol{\beta}_\xi, \mathrm{d}\sigma^2) \Big/ \sum_\xi \int_{\boldsymbol{\beta}_\xi, \sigma^2} L(D_n|\xi, \boldsymbol{\beta}_\xi, \sigma^2)\, \pi(\xi, \mathrm{d}\boldsymbol{\beta}_\xi, \mathrm{d}\sigma^2). \tag{5}$$

2.1.1. Prior specification

For the model ξ, we assume that each predictor has a prior probability $\lambda_n = r_n / P_n$, independent of other predictors, to be included in the model ξ. Further, we impose a constraint on the model size such that $|\xi| \leqslant \bar{r}_n$, where the upper bound $\bar{r}_n$ is prespecified. Then the prior probability of ξ can be expressed as

$$\pi(\xi) \propto \lambda_n^{|\xi|} (1 - \lambda_n)^{P_n - |\xi|}\, I(|\xi| \leqslant \bar{r}_n). \tag{6}$$

The constraint on the model size $|\xi|$ is not crucial to the data analysis. Since the true model is sparse, there is not much loss as long as the size of the true model is less than $\bar{r}_n$. From the perspective of computation, such a constraint facilitates posterior simulations, because, as shown in expression (7), evaluation of the posterior distribution involves inverting a dense matrix of size at most $\bar{r}_n \times \bar{r}_n$. It is known that the computational complexity of inverting such a matrix is $O(\bar{r}_n^3)$. From the perspective of theoretical development, this constraint can be removed by imposing an additional condition, say, there exists $\bar{r}_n$ such that $\pi(|\xi| > \bar{r}_n) \to 0$ with an appropriate rate, without affecting the validity of the following theorems. Refer to Section 7 of Jiang (2007) for more discussions on this issue.

We let the variance σ^2 be subject to an inverse gamma prior distribution with the hyperparameters a_0 and b_0, i.e. $\sigma^2 \sim \mathrm{IG}(a_0, b_0)$. Conditioned on the model ξ and σ^2, we let $\boldsymbol{\beta}_\xi$ be subject to a Gaussian prior:

$$\boldsymbol{\beta}_\xi | \xi, \sigma^2 \sim N_{|\xi|}(0, \sigma^2 V_\xi),$$

where V_ξ is a positive definite matrix depending on ξ.

To facilitate computation, we integrate out $\boldsymbol{\beta}_\xi$ and σ^2 from the posterior distribution (5). The resulting posterior distribution of ξ is given by

$$\pi(\xi|D_n) \propto \left(\frac{r_n}{P_n}\right)^{|\xi|} \left(1 - \frac{r_n}{P_n}\right)^{P_n - |\xi|} \frac{\sqrt{\det(V_\xi^{-1})}}{\sqrt{\det(\mathbf{X}_\xi^{\mathrm{T}} \mathbf{X}_\xi + V_\xi^{-1})}} \times \{2b_0 + \mathbf{y}^{\mathrm{T}}(I - \mathbf{X}_\xi(\mathbf{X}_\xi^{\mathrm{T}}\mathbf{X}_\xi + V_\xi^{-1})^{-1}\mathbf{X}_\xi^{\mathrm{T}})\mathbf{y}\}^{-n/2-a_0}\, I(|\xi| \leqslant \bar{r}_n). \tag{7}$$

For a fully Bayesian variable selection approach, λ_n is usually subject to a beta prior. As pointed out by Scott and Berger (2010), such a prior will induce an automatic multiplicity adjustment for variable selection: the penalty for adding an extra variable increases as P_n increases. The multiplicity control is crucial for variable selection when $P_n > n$. Otherwise, the resulting model tends to be liberal. For example, the BIC places a flat prior on the model space, and the resulting models are overly liberal for high dimensional regression (Broman and Speed, 2002). In this paper, we assume that $\lambda_n \to 0$ as $P_n \to \infty$, which corresponds to condition (13) in theorem 1. This condition provides an automatic mechanism of multiplicity control for the SAM approach.

The posterior (7) is also closely related to the EBIC (Chen and Chen, 2008), which is to choose a model ξ by minimizing the penalized likelihood function

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

$$\mathrm{EBIC}(\xi) = -\hat{l}(\xi) + \frac{|\xi|}{2}\log(n) + \gamma|\xi|\log(P_n),$$

where $\hat{l}(\xi)$ is the log-likelihood function of ξ evaluated at the maximum likelihood estimates of β_ξ and σ^2. If we set $V_\xi^{-1} = \mathbf{X}_\xi^{\mathrm{T}}\mathbf{X}_\xi/n$, $a_0 = b_0 \approx 0$, and $r_n = P_n^{1-\gamma}$ for a constant $\gamma \in (0,1)$, then $\mathbf{X}_\xi^{\mathrm{T}}\mathbf{X}_\xi + V_\xi^{-1} \approx \mathbf{X}_\xi^{\mathrm{T}}\mathbf{X}_\xi$ and $\log(r_n/P_n)^{|\xi|}(1-r_n/P_n)^{P_n-|\xi|} \approx -\gamma|\xi|\log(P_n) + P_n^{1-\gamma}$. Therefore, $\log\{\pi(\xi|D_n)\} \approx -\mathrm{EBIC}(\xi) + C$ for some constant C independent of ξ. A similar approximation between the posterior and the EBIC can be found in Liang *et al.* (2013) for generalized linear models.

2.1.2. Posterior consistency

In what follows, we study the asymptotics of the posterior distribution with respect to the Hellinger distance under appropriate conditions of r_n, $\bar{r}_n$ and V_ξ. For two density functions $f_1(x,y)$ and $f_2(x,y)$, the Hellinger distance is defined as

$$d(f_1, f_2) = \sqrt{\int\int \{f_1^{1/2}(x,y) - f_2^{1/2}(x,y)\}^2 \,\mathrm{d}x\,\mathrm{d}y}.$$

For a matrix Σ, we let $ch_i(\Sigma)$ denote the ith largest eigenvalue, and let $ch_i'(\Sigma)$ denote the ith smallest eigenvalue. Define $B(r_n) = \sup_{\xi:|\xi|=r_n} ch_1(V_\xi^{-1})$, $\bar{B}(r_n) = \sup_{\xi:|\xi|=r_n} ch_1(V_\xi)$, $\tilde{B}_n = \sup_{\xi:|\xi|\leqslant \bar{r}_n} ch_1(V_\xi)$ and $\Delta(p) = \inf_{\xi:|\xi|=p} \Sigma_{i\notin\xi} |\beta_i^*|$.

Theorem 1 follows from Jiang (2007). Here we consider the simple situation that all predictors have been standardized and uniformly bounded, and the sparsity condition (3) holds.

Theorem 1 (posterior consistency). Assume that the data set D_n is drawn from model (1) and all predictors $|x_i| \leqslant 1$. If there is a sequence $\{\epsilon_n\} \in (0,1)$ such that $n\epsilon_n^2 \succ 1$, and the following conditions hold,

$$\Delta(r_n) \prec \epsilon_n, \tag{8}$$

$$\bar{r}_n \ln(1/\epsilon_n^2) \prec n\epsilon_n^2, \tag{9}$$

$$\bar{r}_n \ln(P_n) \prec n\epsilon_n^2, \tag{10}$$

$$\bar{r}_n \ln(n\epsilon_n^2 \tilde{B}_n) \prec n\epsilon_n^2, \tag{11}$$

$$1 \leqslant r_n \leqslant \bar{r}_n < P_n, \tag{12}$$

$$r_n \prec P_n, \tag{13}$$

$$B(r_n) \prec n\epsilon_n^2, \tag{14}$$

then the posterior consistency holds, i.e. there is a constant $c_1 > 0$ such that

$$P^*[\pi\{d(f, f^*) > \epsilon_n | D_n\} > \exp(-c_1 n\epsilon_n^2)] < \exp(-c_1 n\epsilon_n^2), \tag{15}$$

for sufficiently large n, where the Hellinger distance is

$$d^2(f, f^*) = \int\int |\phi^{1/2}(y; x^{\mathrm{T}}\beta^*, \sigma^*) - \phi^{1/2}(y; x^{\mathrm{T}}\beta, \sigma)|^2 \,\nu_x(\mathrm{d}x)\,\mathrm{d}y,$$

f^* is as defined in expression (2), f is the density function for a model simulated from the posterior and P^* denotes the probability measure of data generation.

The proof of theorem 1 can be found in the on-line supplementary material. The validity of this theorem depends on the choice of the prior covariance matrix V_ξ or, more precisely, the

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

largest eigenvalues of V_ξ and V_ξ^{-1}. For most choices of V_ξ, $B(r_n) \leqslant Br_n^v$ and $\tilde{B}_n \leqslant Br_n^v$ hold for some positive constants B and v. For example, if $V_\xi = cI_{|\xi|}$, where I_p is an identity matrix of order p, then both $B(r_n)$ and $\tilde{B}_n$ are constants. One can also set $V_\xi^{-1} = c\,E(x_\xi^{\mathrm{T}} x_\xi)$ as in Zellner (1986) and Liang *et al*. (2008), where the expectation is replaced by its sample empirical estimator. If all x_is are standardized with zero mean and equal variance, then $B(r_n) \leqslant r_n$. Furthermore, if x_i has some specific covariance structure, then $\tilde{B}_n$ can be shown to be bounded. For example, if all pairwise covariances $\mathrm{cov}(x_i, x_j) = \rho \in (0, 1)$, then $\tilde{B}_n = 1 - \rho$; if x_i is generated from a finite order auto-regressive or moving average model without root for the characteristic polynomial on the unit circle, then $\tilde{B}_n$ is also bounded by some constant (Bickel and Levina, 2004). In general, for standardized x_is, we can choose a 'ridge prior' as suggested by Gupta and Ibrahim (2009): let $V_\xi^{-1} = E(x_\xi^{\mathrm{T}} x_\xi) + c_n I_{|\xi|}$; then $B(r_n) \leqslant r_n + c_n$ and $\tilde{B}_n \leqslant c_n^{-1}$, where c_n is proportional to $1/n$, i.e. $c_n \propto n^{-1}$.

Theorem 1 assumes that all the predictors are uniformly bounded. However, this condition might not be satisfied for continuous predictors. In what follows, we relax this condition.

Condition 1. There are some constants $M > 0$ and $\delta > 0$ such that for any subvector of x, $(x_{j_1}, \ldots, x_{j_r})^{\mathrm{T}}$ with $1 \leqslant j_1 < \ldots < j_r \leqslant P_n$, $r \leqslant \bar{r}_n$, and $|a_i| \leqslant \delta$ for $i = 1, \ldots, r$, we have

$$E\left[\exp\left\{\left(\sum_{i=1}^{r} a_i x_{j_i}/r\right)^2\right\}\right] \leqslant \exp(M),$$

where the expectation is taken with respect to the distribution of the random variables x_{j_i}.

Condition 1 controls the tail distribution of x such that the marginal distribution of y does not change dramatically in the sense of d_1-divergence (which is defined in the on-line supplementary material) with respect to the change of $\boldsymbol{\beta}$. If x has a heavy tail distribution, the prior information around f^* will be too weak to induce good posterior asymptotics. If all the variables are bounded by some constant m, as assumed in theorem 1, then

$$E\left[\exp\left\{\left(\sum_{i=1}^{r} a_i x_{j_i}/r\right)^2\right\}\right] \leqslant \exp(\delta^2 m)$$

and thus condition 1 holds. If x follows a Gaussian process with zero mean and $\mathrm{var}(x_i) = 1$, then $(\Sigma_{i=1}^{r} a_i x_{j_i})^2$ follows the $s\chi_1^2$-distribution, where the scale $s \leqslant \delta^2 r^2$. Therefore, by the moment-generating function of the χ_1^2-distribution,

$$E\left[\exp\left\{\left(\sum_{i=1}^{r} a_i x_{j_i}\right)^2 \Big/ r^2\right\}\right] \leqslant (1 - 2\delta^2)^{-1/2},$$

condition 1 holds.

Under condition 1, we have the following theorem whose proof can be found in the on-line supplementary material.

Theorem 2. Assume that the data set D_n is generated from model (1) and all the predictors satisfy condition 1. Given a sequence $\{\epsilon_n\}$, $\epsilon_n \to 0$, and $n\epsilon_n^2 \to \infty$, if the conditions (9)–(14) of theorem 1 hold, the model is strictly sparse, i.e. condition (4) holds, and $r_n \succ 1$, then the posterior consistency still holds, i.e. there is a constant $c_1 > 0$ such that, for sufficiently large n,

$$P^*[\pi\{d(f, f^*) > \epsilon_n | D_n\} > \exp(-c_1 n\epsilon_n^2)] < \exp(-c_1 n\epsilon_n^2).$$

Note that condition (8) of theorem 1 is redundant here as, for large n, $r_n > |\mathbf{t}|$ and $\Delta(r_n) = 0$

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

hold under the sparseness condition (4). The following corollary concerns the convergence rate of the posterior distribution.

Corollary 1. Consider a strict sparse model, i.e. condition (4) holds. Suppose that $P_n < \exp(Cn^{\alpha})$ for some $\alpha \in (0,1)$, and the prior specification in Section 2.1.1 is used such that $B(r_n) \leqslant Br_n^v$ and $\tilde{B}_n \leqslant Bn^v$ for some positive constants B and v. Take

$$r_n < \bar{r}_n \prec \log^k(n),$$

or

$$r_n < \bar{r}_n \prec n^q,$$

for some $k > 0$ or some $0 < q < \min\{1-\alpha, 1/v\}$. Then the convergence rate in theorem 2 can be taken as

$$\epsilon_n = O\{n^{-(1-\alpha)/2} \log^{k/2}(n)\},$$

or

$$\epsilon_n = O(n^{-(\min\{1-\alpha-q, 1-vq\})/2}).$$

2.1.3. Remark for condition 1

Condition 1 imposes a stringent requirement for the distribution of predictors, although one may treat each predictor as bounded in practice. If the distribution of a predictor is heavy tailed, then condition 1 fails. To address this issue, a much weaker condition is proposed below.

Condition 2. There are some constants $M > 0$ and $\delta > 0$ such that, for any subvector of x, $(x_{j_1}, \ldots, x_{j_r})^{\mathrm{T}}$ with $1 \leqslant j_1 < \ldots < j_r \leqslant P_n$, $r \leqslant \bar{r}_n$, and $|a_i| \leqslant \delta$ for $i = 1, \ldots, r$, we have

$$E\left(\sum_{i=1}^{r} a_i x_{j_i}/r\right)^2 \leqslant M.$$

Condition 2 imposes conditions on only the second moments of predictors. If all predictors have uniformly bounded second moments, then condition 2 holds. It is shown in the on-line supplementary material that, if condition 2 holds, a slightly weaker result of theorem 2 can be obtained as follows:

$$\lim_{n\to\infty} P^*[\pi\{d(f, f^*) > \epsilon_n | D_n\} > \exp(-c_1 n\epsilon_n^2)] = 0. \tag{16}$$

Under condition 1, $\pi\{d(f, f^*) > \epsilon_n | D_n\}$ converges in L_1 exponentially, whereas result (16) implies only convergence in probability.

2.2. Variable screening and selection for correctly specified models

To serve the purpose of variable selection for the case that the true model $\mathbf{t}$ exists, i.e. condition (4) holds, we need a consistent variable selection procedure. In this paper, we consider two procedures, which are based on the marginal inclusion probability and MAP model. The former is described below, and the latter is described in the on-line supplementary material.

Variable selection by marginal inclusion probability has been used in high dimensional Bayesian analysis; see for example Barbieri and Berger (2004) and Liang *et al.* (2013). In this paper, we study the property of marginal inclusion probability based on the posterior convergence result (15) or (16), which implies that $\pi\{d(f, f^*) > \epsilon_n | D_n\} \to^{\mathrm{p}} 0$. Since our objective is to recover all the true predictors, we need to explore the relationship between the distance in

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

distributions and the difference in models. Let q_j denote the marginal inclusion probability of the predictor $\mathbf{x}_j$, which is given by

$$q_j = \pi(j \in \xi | D_n) = \sum_{\xi \in \{\tilde{\xi}: j \in \tilde{\xi}\}} \pi(\xi | D_n).$$

Naturally we expect that a true predictor has a high marginal inclusion probability and a false predictor has a low marginal inclusion probability. However, this might not be true in the high dimensional scenario where multicollinearity is present. Wilson (2010) studied the relationship between low rankness of the design matrix and consistency of pairwise model Bayes factors and global posterior probabilities. If a true predictor can be well approximated or even replaced by a couple of spurious predictors, then the marginal inclusion probabilities of the spurious predictors are inflated. Therefore, to select true predictors consistently, it is necessary to control the severeness of multicollinearity. For this purpose, we introduce the following identifiability condition which controls the severeness of multicollinearity through the sequence $\{\epsilon_n\}$.

Condition 3. A predictor $\mathbf{x}_k$ is said to be identifiable among all other predictors, if, for any $1 \leqslant j_1, \ldots, j_{\bar{r}_n} \leqslant P_n$ ($j_i \neq k$ for all i) and $b_i \in \mathbb{R}$,

$$E\left[\exp\left\{-\left(x_k + \sum_{i=1}^{\bar{r}_n} b_i x_{j_i}\right)^2\right\}\right] \leqslant 1 - \delta_n, \qquad \delta_n \succ \epsilon_n^2,$$

where x_l denotes a generic observation of the predictor $\mathbf{x}_l$.

This condition states that, if a predictor $\mathbf{x}_k$ is identifiable, then there does not exist a linear combination of other predictors in $\mathbf{X}$ which can mimic it.

If all the predictors have mean 0 and are bounded with $|x_i| \leqslant 1$, then

$$\begin{aligned} E\left[\exp\left\{-\left(x_k + \sum_{i=1}^{\bar{r}_n} b_i x_{j_i}\right)^2\right\}\right] &\leqslant E\left[1 - \frac{1 - \exp\left\{-\left(1 + \sum_i |b_i|\right)^2\right\}}{\left(1 + \sum_i |b_i|\right)^2}\left(x_k + \sum_i b_i x_{j_i}\right)^2\right] \\ &\leqslant 1 - \{1 - \exp(-1)\}\frac{\operatorname{var}\left(x_k + \sum_i b_i x_{j_i}\right)}{(1 + \sum |b_i|)^2} \\ &\leqslant 1 - \frac{1 - \exp(-1)}{(\bar{r}_n + 1)} c h_1'\{\operatorname{var}(\tilde{x})\}, \end{aligned}$$

where $\tilde{x} = (x_k, x_{j_1}, \ldots, x_{j_{\bar{r}_n}})^{\mathrm{T}}$, and $c h_1'\{\operatorname{var}(\tilde{x})\}$ is the smallest eigenvalue of the covariance matrix of $\tilde{x}$. Hence, if $c h_1'\{\operatorname{var}(\tilde{x})\} \succ \bar{r}_n \varepsilon_n^2$ for any choice of $\tilde{x}$, x_t is identifiable. Furthermore, if $c h_1'\{\operatorname{var}(x)\} \succ \bar{r}_n \epsilon_n^2$, then all the predictors are identifiable, where $x = (x_1, \ldots, x_{P_n})^{\mathrm{T}}$ denotes a generic observation of all predictors.

If all the predictors follow the standard normal distribution, by the moment-generating function of the χ^2-distribution, we have

$$\begin{aligned} E[\exp\{-(x_k + \textstyle\sum b_i x_{j_i})^2\}] &= \left\{1 + 2\operatorname{var}\left(x_k + \sum_i b_i x_{j_i}\right)\right\}^{-1/2} \\ &< \max\left\{0.5, 1 - \operatorname{var}\left(x_k + \sum_i b_i x_{j_i}\right)/8\right\}. \end{aligned}$$

Thus, if $\operatorname{var}(x_k + \Sigma_i b_i x_{j_i}) \succ \epsilon_n^2$, then the identifiability condition is satisfied. Since

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

$$\mathrm{var}\left(x_k + \sum_i b_i x_{j_i}\right) \geqslant \mathrm{var}(x_k|\tilde{x}\backslash x_k) \geqslant c\,h_1'\{\mathrm{var}(\tilde{x})\} \geqslant c\,h_1'\{\mathrm{var}(x)\},$$

by the same arguments, all predictors are identifiable if $c\,h_1'\{\mathrm{var}(x)\} \succ \epsilon_n^2$. Further, it can be shown (in the proof of theorem 3) that, if a true predictor $\mathbf{x}_t$ is identifiable, then $d(f^*, f) \geqslant \epsilon_n$ for any density f that does not select $\mathbf{x}_t$. Therefore, we have the following sure screening property.

Theorem 3 (sure screening). Assume that all the conditions of theorem 2 hold. If a true predictor $\mathbf{x}_t$ ($t \in \mathbf{t}$) is identifiable, then $P^*[\pi\{t \in \xi|D_n\} < 1 - \exp(-c_1 n\epsilon_n^2)] < \exp(-c_1 n\epsilon_n^2)$, where ξ denotes a model sampled from the posterior distribution. Furthermore, if condition 2 holds, then the weaker convergence holds: $\lim_n P^*[\pi\{t \in \xi|D_n\} < 1 - \exp(-c_1 n\epsilon_n^2)] = 0$.

The proof of theorem 3 can be found in the on-line supplementary material. Let $\xi_q = \{i : 1 \leqslant i \leqslant P_n, q_i > q\}$ denote a model including all the predictors with the marginal inclusion probability greater than a threshold value of $q \in (0, 1)$. Then it follows from theorem 3 that $\pi(\mathbf{t} \subset \xi_q) \to^{\mathrm{p}} 1$ if all true predictors are identifiable. To determine the threshold value q, we adopt the multiple-hypothesis testing procedure that was used in Liang *et al*. (2013), where it is used for selecting variables for a logistic regression model. The procedure was originally proposed in Liang and Zhang (2008). Compared with other false discovery rate (FDR) control procedures, one advantage of this procedure is that it is applicable under general dependence between marginal inclusion probabilities.

Theorem S in the on-line supplementary material establishes the consistency of MAP models. Thus, we have two criteria for variable selection, by either marginal inclusion probabilities or the MAP model. Although theorem 3 states only the sure screening property (the sensitivity is asymptotically 1, but the specificity is not controlled), we suggest the use of marginal inclusion probabilities as the criterion for variable selection. Under this criterion, one may select more predictors, which are worthy of further investigation through a possibly different and expensive experiment. Since the sample size n can be small for a real problem, the MAP model might not capture all true predictors although it is consistent in theory.

2.3. Variable screening for misspecified models

Let $D_n^{\mathbf{s}} = \{\mathbf{y}, \mathbf{X}_{\mathbf{s}}\}$ denote a fixed subset of observations, where $\mathbf{s} \subset \{1, 2, \ldots, P_n\}$ and $\mathbf{X}_{\mathbf{s}}$ contains only $s = |\mathbf{s}| < P_n$ predictors with the indices belonging to $\mathbf{s}$. Assume that $\mathbf{X}_{\mathbf{s}}$ does not include all of the true predictors of model (1). Therefore, the model

$$\mathbf{y} = \mathbf{X}_{\mathbf{s}}\boldsymbol{\beta}_{\mathbf{s}} + \boldsymbol{\varepsilon}$$

is misspecified. Under the Bayesian framework, some work for misspecified models has been done by Berk (1966), Kleijn and van der Vaart (2006), Shalizi (2009) and De Blasi and Walker (2013), among others. Let $\mathcal{P}_s$ denote a set of parameterized densities for all possible models formed with $\mathbf{X}_{\mathbf{s}}$. It is obvious that $f^* \notin \mathcal{P}_s$. Let f_0 denote the minimization point of the Kullback–Leibler divergence in $\mathcal{P}_s$, i.e.

$$f_0 = \arg\min_{f \in \mathcal{P}_s} \int \ln(f^*/f) f^*. \tag{17}$$

With a slight abuse of notation, we use, as in previous sections, ξ to denote a model selected among the predictors in $\mathbf{X}_{\mathbf{s}}$. Further, the same prior distributions are specified for the parameters σ^2 and $\boldsymbol{\beta}_\xi$, i.e.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

$$\pi(\sigma^2) \sim \mathrm{IG}(a_0, b_0),$$

$$\pi(\boldsymbol{\beta}_\xi | \xi, \sigma^2) \sim N(0, \sigma^2 V_\xi).$$

Under these priors, we show in theorem 4 that the density f_0 can be consistently estimated by the models sampled from the posterior $\pi(\xi | D_n^{\mathbf{s}})$ as $n \to \infty$. To prove theorem 4, $\pi(\xi)$ does not need to be specified, which is only required to be positive for all possible models, i.e. $\pi(\xi) > 0$ for all ξ.

Parallel to condition 1, we introduce the following condition regarding the range of x_is.

Condition 4. There is a constant $\delta > 0$ such that for any $\mathbf{a} = (a_1, \ldots, a_s)^{\mathrm{T}} \in \mathbb{R}^s$, with $|a_i| \leqslant \delta$ for $i = 1, \ldots, s$, we have

$$E\{(\mathbf{a}^{\mathrm{T}} x_{\mathbf{s}})^2\} < \infty,$$

where $x_{\mathbf{s}}$ denote a generic observation of the predictors included in $\mathbf{X}_{\mathbf{s}}$.

Theorem 4. (posterior consistency for misspecified models). Assume that condition 4 holds for a given subset $\mathbf{s}$. Under the prior setting as described above, for any $\epsilon > 0$,

$$\pi[\{f \in \mathcal{P}_s : d(f_0, f) > \epsilon\} | D_n^{\mathbf{s}}] \to 0, \qquad \text{almost surely,} \tag{18}$$

as $n \to \infty$, where f_0 is as defined in equation (17), and f is a parameterized density proposed from posterior.

It follows from lemma 1.2 (in the on-line supplementary material) that $f_0 = \phi(y; \boldsymbol{\beta}_0^{\mathrm{T}} x_{\mathbf{s}}, \sigma_0)\, \nu(x_{\mathbf{s}})$, where $\nu(x_{\mathbf{s}})$ denotes the probability measure of $x_{\mathbf{s}}$, and $\boldsymbol{\beta}_0$ is given by

$$\boldsymbol{\beta}_0 = \arg\min_{\boldsymbol{\beta}_{\mathbf{s}}} E(\boldsymbol{\beta}_{\mathbf{s}}^{\mathrm{T}} x_{\mathbf{s}} - \boldsymbol{\beta}^{*\mathrm{T}} x)^2,$$

i.e. $\boldsymbol{\beta}_0^{\mathrm{T}} x_{\mathbf{s}}$ is the projection of $\boldsymbol{\beta}^{*\mathrm{T}} x$. Let $\tilde{\mathbf{s}}$ denote the subset of $\mathbf{s}$ corresponding to non-zero entries of $\boldsymbol{\beta}_0$. Following the proof of theorem 3, we have the following corollary.

Corollary 2. Assume that the conditions of theorem 4 hold. If $x_{\mathbf{s}}$ does not have exact multicollinearity between variables, i.e. there does not exist a non-zero vector $\mathbf{a} \in \mathbb{R}^s$ such that $P(\mathbf{a}^{\mathrm{T}} x_{\mathbf{s}} = 0) = 1$, then for the posterior probability of model ξ, conditioned on the subset $D_n^{\mathbf{s}}$, we have

$$\pi(\tilde{\mathbf{s}} \subset \xi | D_n^{\mathbf{s}}) \xrightarrow{\mathrm{p}} 1.$$

Corollary 2 implies that the marginal inclusion probability criterion is still applicable for selection of predictors in $\mathbf{X}_{\mathbf{s}}$. For any true predictor $\mathbf{x}_t$ with $t \in \mathbf{t} \cap \mathbf{s}$, it will be asymptotically selected under this criterion if and only if $\beta_{0,t} \neq 0$, where $\beta_{0,t}$ is the entry of $\boldsymbol{\beta}_0$ corresponding to $\mathbf{x}_t$. Let y denote a generic observation of the response variable, let $x_{\mathbf{t}}$ denote a generic observation of the true predictors and let $\boldsymbol{\beta}_{\mathbf{t}}^*$ denote the regression coefficient vector of the true predictors. Since

$$\boldsymbol{\beta}_0 = \Sigma_{\mathbf{s}}^{-1} \Sigma_{\mathbf{s},\mathbf{t}} \boldsymbol{\beta}_{\mathbf{t}}^* = \Sigma_{\mathbf{s}}^{-1} \mathrm{cov}(x_{\mathbf{s}}, y), \tag{19}$$

where $\Sigma_{\mathbf{s}} = \mathrm{var}(x_{\mathbf{s}})$ and $\Sigma_{\mathbf{s},\mathbf{t}} = \mathrm{cov}(x_{\mathbf{s}}, x_{\mathbf{t}})$, $\tilde{\mathbf{s}}$ is determined by the correlation structure of $x_{\mathbf{s}}$ and $x_{\mathbf{t}}$ and the value of the true regression coefficients. However, when $\mathbf{t} \not\subseteq \mathbf{s}$, it is unclear whether $\beta_{0,t}$ is exactly 0 or not. Following from equation (19), if $\mathrm{cov}(x_{\mathbf{s}}, y) \neq 0$, then $\boldsymbol{\beta}_0 \neq 0$ and thus at least one predictor in $\mathbf{s}$ will be selected according to the marginal inclusion probability criterion. This motivates us to propose an iterative variable selection method for the subset data, provided that the following condition holds.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

Condition 5. $|\mathrm{cov}(x_t, y)| > 0$ for any true predictor $\mathbf{x}_t$.

The iterative variable selection method can be described as follows: first select predictors from $\mathbf{X_s}$ based on the marginal inclusion probabilities estimated from the posterior $\pi(\xi|D_n^{\mathbf{s}})$; denote the set of unselected predictors by $\mathbf{X_{s'}}$, and then select predictors again from $\mathbf{X_{s'}}$ based on the marginal inclusion probabilities estimated from the posterior $\pi(\xi|\mathbf{y}, \mathbf{X_{s'}})$; keep running the iterative procedure until no predictors can be selected. Denote all the selected predictors by $\mathbf{X_{\tilde{s}}}$. Under assumption 5, this procedure ensures that all true predictors that are contained in $\mathbf{X_s}$ will be selected.

There are a few remarks on this iterative procedure.

(a) Other than true predictors, how many other predictors will be selected? Generally speaking, all the predictors that are correlated with $\mathbf{X_t}$ will be selected, because they are correlated with $\mathbf{y}$, where $\mathbf{X_t}$ denotes the set of all true predictors in D_n. This means that, if all the predictors are linearly correlated, it is futile to apply this iterative procedure, as all the predictors will be selected. The rationale underlying the SAM strategy lies in the belief that, for a big data set, there are only a small proportion of predictors that are linearly correlated with $\mathbf{X_t}$. It is easy to show that if there are p_n predictors correlated with the true predictor $\mathbf{X_t}$, $p_n \prec P_n$, then there are at most cp_n predictors to be selected into $\mathbf{X_{\tilde{s}}}$ for some constant c.

(b) To ensure that all true predictors in $\mathbf{X_s}$ are selected, a liberal way, as we proposed, is to repeat the selection iteratively until no predictors can be selected. In practice, it is rare to have the case that $\beta_{0,t}$ is 0 exactly. Hence, the iterative procedure is not always necessary, and one may set an upper bound for the iteration number. In this paper, we set the iteration number to 2.

(c) Theorem 4 requires the prior probability $\pi(\xi) > 0$ for all possible models. Hence, the independent prior $\pi(\xi) = \lambda_{\mathbf{s}}^{|\xi|}(1-\lambda_{\mathbf{s}})^{|\mathbf{s}|-|\xi|}$ for some $\lambda_{\mathbf{s}} \in (0,1)$ is still applicable here. When $|\mathbf{s}|$ is large, we suggest imposing an upper bound for the model size to avoid expensive computation caused by inverting high order matrices. Then the posterior should asymptotically concentrate on the models with the density function minimizing the Kullback–Leibler divergence under the constraint of model size. In this case, although corollary 2 does not hold anymore, the marginal inclusion probabilities for correlated and uncorrelated predictors are distinct and the sure screening property of the marginal inclusion probability criterion should still hold.

(d) For data splitting, one extreme choice is $s = 1$. In this case, the SAM method will perform like SIS (Fan and Lv, 2008) with only the marginal utility being used in variable screening. However, this choice of s is not recommended, as the SAM approach tends to work better with a large value of s, say $s \geqslant 500$. A large value of s allows the joint information of predictors to be used in variable screening, and hence the SAM approach has a better chance of screening out the predictors that are spuriously correlated with the response than SIS. Since a large model is always penalized in either stage of the SAM approach, the marginal inclusion probabilities of the spuriously correlated predictors tend to be low. It is often to have the case that $|\mathrm{corr}(\mathbf{x}_i, \mathbf{y})| > |\mathrm{corr}(\mathbf{x}_j, \mathbf{y})|$, where $\mathrm{corr}(\cdot,\cdot)$ denotes the correlation function, whereas the marginal inclusion probability of $\mathbf{x}_i$ is lower than that of $\mathbf{x}_j$. This is because $\mathbf{x}_i$ is highly collinear with a true predictor, say $\mathbf{x}_k$, and $\mathbf{x}_j$ is complementary to $\mathbf{x}_k$ for the regression. This case distinguishes the SAM approach from SIS. In addition, a large value of s will improve the accuracy of the marginal inclusion probability-based variable screening procedure, which, as stated before, is multiple-hypothesis test based.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

3. Split-and-merge approach and its implementation

3.1. Split-and-merge approach

Based on the priors given in Section 2, the SAM approach can be summarized as follows.

(a) Split P_n predictors into K_n groups, $\mathbf{s}_1, \ldots, \mathbf{s}_{K_n}$, with $\max_{i=1,\ldots,K_n} |\mathbf{s}_i| \leqslant s$ for a prespecified value of s.

(b) *Stage 1*: apply the iterative procedure, proposed in Section 2.3, to select predictors from each subset $D_n^{\mathbf{s}_i} = \{\mathbf{y}, \mathbf{X}_{\mathbf{s}_i}\}$, $i = 1, \ldots, K_n$, at an FDR level of α_1. At each iteration, the Bayesian variable selection is subject to the prior setting $\pi(\sigma) \sim \mathrm{IG}(a_0, b_0)$, $\pi(\xi) = \lambda_{\mathbf{s}}^{|\xi|}(1-\lambda_{\mathbf{s}})^{|\mathbf{s}|-|\xi|} I(|\xi| \leqslant \bar{s})$ and $\pi(\boldsymbol{\beta}_\xi | \xi, \sigma) \sim N\{0, \sigma^2(\mathbf{X}_\xi^{\mathrm{T}}\mathbf{X}_\xi + \tau I)/n\}$, where $\bar{s}$ denotes the upper bound of the size of the models that are considered for each subset of data. Let $\tilde{\mathbf{s}}_1, \ldots, \tilde{\mathbf{s}}_{K_n}$ denote the sets of indices of the selected predictors from the K_n subsets.

(c) *Stage 2*: merge the sets $\tilde{\mathbf{s}}_1, \ldots, \tilde{\mathbf{s}}_{K_n}$ into a single set $\tilde{\mathbf{S}} = \cup_{i=1}^{K_n} \tilde{\mathbf{s}}_i$, and define $p_n = |\tilde{\mathbf{S}}|$. Perform Bayesian variable selection on the subset data $D_n^{\tilde{\mathbf{S}}} = \{\mathbf{y}, \mathbf{X}_{\tilde{\mathbf{S}}}\}$ at an FDR level of α_2. The Bayesian variable selection is subject to the prior setting $\pi(\sigma) \sim \mathrm{IG}(a_0, b_0)$, $\pi(\xi) = (r_n/p_n)^{|\xi|}(1-r_n/p_n)^{p_n-|\xi|} I(|\xi| \leqslant \bar{r}_n)$ and $\pi(\boldsymbol{\beta}_\xi | \xi, \sigma) \sim N\{0, \sigma^2(\mathbf{X}_\xi^{\mathrm{T}}\mathbf{X}_\xi + \tau I)/n\}$.

Clearly, it follows from the theory that was developed in Section 2 that the SAM approach will lead to a consistent selection of true predictors. In step 3, the predictors can also be selected according to the MAP model.

3.2. Simulation and hyperparameter setting

Since different predictors can be highly correlated, the posterior (7) can have a rugged energy landscape for some problems. Conventional MCMC algorithms, such as the Metropolis–Hasting algorithm, tend to become trapped in a local mode in simulations. To avoid this problem, the stochastic approximation Monte Carlo (SAMC) algorithm (Liang *et al.*, 2007) was used in this paper. The SAMC algorithm belongs to the class of adaptive MCMC algorithms. However, unlike conventional adaptive MCMC algorithms, e.g. the adaptive Metropolis algorithm (Haario *et al.*, 2001), which adapt the proposal distribution, the SAMC algorithm adapts the invariant distribution at each iteration. As explained in Liang *et al.* (2007), the self-adjusting mechanism of the invariant distribution enables the SAMC algorithm to be immune to local trap problems. As shown in Liang (2009), the SAMC algorithm is essentially a dynamic importance sampling algorithm, and quantities of interest can be estimated by averaging the samples with respect to their importance weights. Refer to the on-line supplementary material for the details of implementation of this algorithm.

Another issue related to the implementation of the SAM approach is how to set hyperparameters. The prior distribution of the SAM approach contains an important hyperparameter, namely r_n. According to theorem 2 and theorem S in the on-line supplementary material, r_n can slowly grow with P_n for a good convergence rate of the posterior and the MAP model. In Section 2.1.1, we establish the connection between the posterior (7) and the EBIC by choosing $r_n = P_n^{1-\gamma}$. Chen and Chen (2008) showed that a choice of $\gamma > 0.5$ is preferred for high dimensional regression. In Liang *et al.* (2013), for a similar hyperparameter, Chen and Chen (2010) suggested the rule

$$\gamma = \inf\{\tilde{\gamma} : \arg\max_{|\xi|} \pi(|\xi||D_n, \tilde{\gamma}) = |\xi_{\mathrm{MAP}, \tilde{\gamma}}|\}, \tag{20}$$

i.e. choosing the minimum value of γ such that the mode of $\pi(|\xi||D_n)$ coincides with the size of the MAP model ξ_{MAP}. If the resulting value of γ is greater than or equal to 1, truncate it

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

to a value less than 1, say 0.99. We can also apply this rule to determine the value of γ and set $r_n = P_n^{1-\gamma}$. In practice, one may try a sequence of γ-values and then choose the smallest for which the mode of the posterior of model size coincides with the size of the MAP model. Our experience shows that, when n is not too small and $\log(P_n)/n$ is not too large, the choice of γ does not affect the result very much. In this paper, we set $\gamma = 0.75$ and $r_n = P_n^{1-\gamma}$ unless otherwise stated. To make the prior for the subset data regression compatible with the prior for the whole data set regression, we set $\lambda_{\mathbf{s}} = |\mathbf{s}|^{-\gamma}$. For other hyperparameters, we set $a_0 = b_0 = 1$ and $\tau = 0.01$.

In addition to r_n, the SAM approach contains a few more free parameters, including $\bar{s}$, $\bar{r}_n$, α_1 and α_2. In practice, we usually set $\bar{s}$ and $\bar{r}_n$ such that the probabilities of the models with size exceeding them are negligible. We usually set α_1 to a large value, e.g. 15% or 10%, to minimize the chance of losing true predictors. In general, a large value of α_1 should be used for a small subset and a small value of α_1 for a large subset, as a large subset will provide a more accurate estimate of the FDR. As in conventional multiple-hypothesis tests, the choice of α_2 may largely affect the results of the SAM approach especially when n is small. The value of α_2 may be determined by using a test data set or through a cross-validation procedure. For simplicity, we set it to 1% in this paper. When n is large, as in Liang *et al.* (2013), we can show under posterior consistency that the marginal inclusion probability tends to be dichotomized, i.e. the marginal inclusion probabilities of the true and false predictors tend to 1 and 0 respectively. Hence, the choices of α_1 and α_2 are not crucial to the results of the SAM approach especially when n is large.

4. Simulated examples

4.1. Toy examples

4.1.1. Example 1

This example confirms the theoretical results that were established in theorem 3 and theorem S in the on-line supplementary material. For this purpose, the predictors are directly selected without data splitting.

This example consists of multiple data sets with different values of n ranging from 20 to 120. For each value of n, we set $P_n = n^{1.5}$ and simulated 100 data sets independently. For each data set, the design matrix $\mathbf{X}$ was generated from a multivariate normal distribution. The variance of each column of $\mathbf{X}$ was set to be 1, and the correlation coefficient between different columns of $\mathbf{X}$ was set to be 0.25, which represents a strong correlation for real gene expression data (see for example Johnson and Rossel (2012)). For each data set, we set $\sigma^* = 1$ and chose the first five columns of $\mathbf{X}$ as the true predictors with the regression coefficients being 0.7, 0.9, 1.1, 1.3 and 1.5.

For each data set, the SAMC algorithm was run for $270P_n$ iterations, where the first $20P_n$ were for the burn-in process. In the simulations, we set the prior hyperparameters $r_n = \sqrt{n}$ and $\bar{r}_n = 2\sqrt{n}$, and set the gain factor sequence as, with $k_0 = 10P_n$,

$$a_k = k_0/\max\{k_0, k\}, \tag{21}$$

where k indexes the number of iterations of the SAMC algorithm.

Fig. 1 shows the distributions of marginal inclusion probabilities for the five true predictors and one false predictor. It is easy to see that, as n increases, the marginal inclusion probabilities of the true predictors converge towards 1, whereas the marginal inclusion probability of the false predictor stays close to 0. This confirms the sure screening property of the marginal inclusion probability as established in theorem 3. Fig. 2 shows that the probability that the MAP model catches the true model increases with n. This confirms theorem S in the on-line supplementary material that the MAP model is consistent. A comparison of Fig. 1 and Fig. 2 shows that, from

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

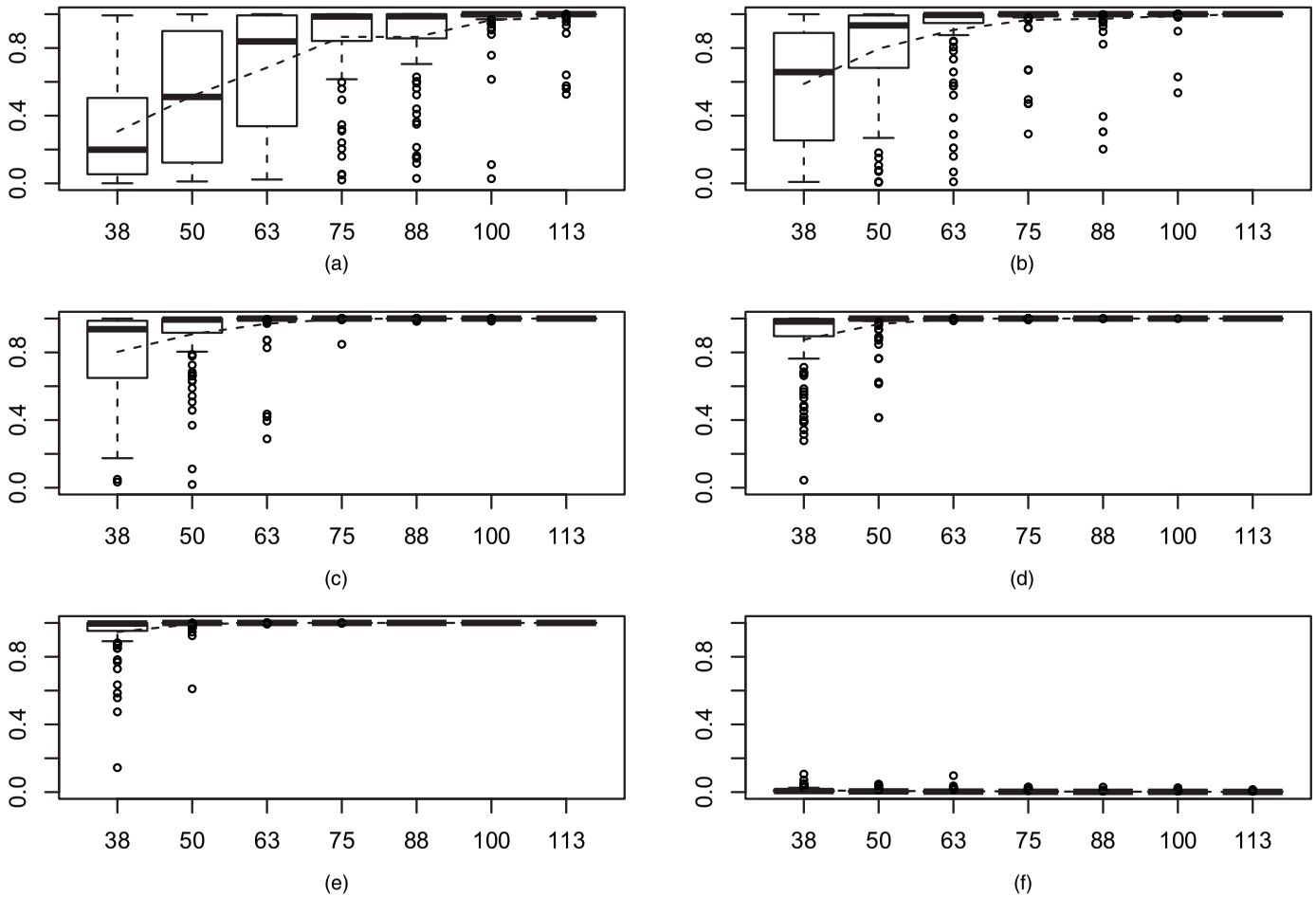


Fig. 1. Marginal inclusion probabilities of six predictors for the case of example 1 that the mutual correlation coefficient between different predictors is 0.25 for true regression coefficients (a) 0.7, (b) 0.9, (c) 1.1, (d) 1.3, (e) 1.5 and (f) 0: in each plot the seven boxplots are for sample sizes $n = 38, 50, 63, 75, 88, 100, 113$; each boxplot shows the distribution of the marginal inclusion probabilities of one predictor calculated from 100 simulated data sets; ------, mean value of the marginal inclusion probabilities

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

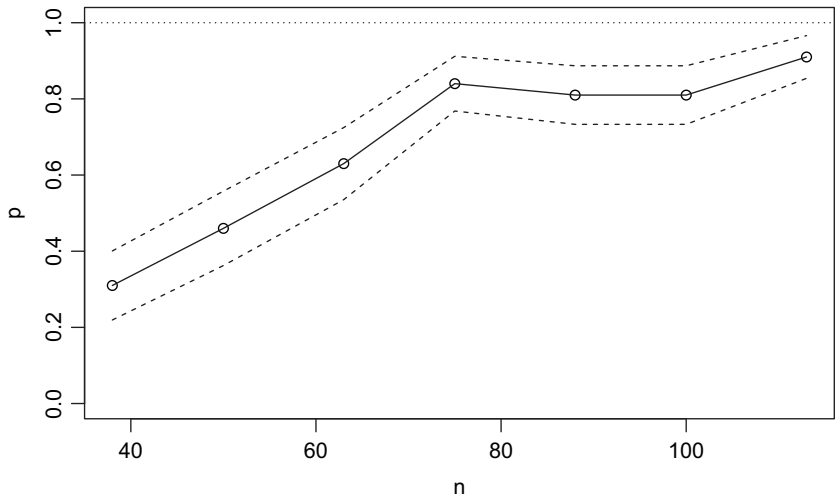


Fig. 2. Plot of the probability that the MAP model coincides with the true model (------, 95% confidence interval of the probability): for each value of n, the probability was estimated from 100 simulated data sets

the perspective of variable selection, the marginal inclusion probability may work better than the MAP model, as the former seems to converge faster than the latter.

Later, a simulation study was conducted to study the stability of the SAM approach when the multicollinearity is extremely strong. We let the pairwise correlation increase to 0.9 and let n increase from 30 to 125, while keeping other settings unchanged. With such extremely high multicollinearity, the MAP criterion almost never catches the true model. However, as shown in Fig. 3, there is still a clear trend that the marginal inclusion probabilities of the true predictors increase with the sample size n.

4.1.2. Example 2

This example illustrates the SAM approach. It consists of 100 simulated data sets, each consisting of $n = 150$ observations and $P_n = 1000$ predictors. For each data set, the design matrix $\mathbf{X}$ was generated from a multivariate normal distribution. The first 100 columns of $\mathbf{X}$ are mutually correlated with an equal correlation coefficient of 0.25, and independent of the remaining 900 columns. The remaining 900 columns are mutually independent. The first three columns were chosen as the true predictors with the regression coefficients being 1.5, 3.0 and 4.5.

We randomly split each data set into 50 subsets with $s = 20$. In stage 1, the predictors are iteratively selected twice for each subset. Each run of the SAMC algorithm consists of $270|\mathbf{s}|$ iterations, where the first $20|\mathbf{s}|$ iterations were for the burn-in process, and $\mathbf{s}$ can be a subset directly split from the full data set or a remainder of a subset containing only unselected predictors. The gain factor sequence was set as $a_k = 10|\mathbf{s}|/\max\{10|\mathbf{s}|, k\}$, where k indexes the number of iterations of the SAMC algorithm. The prior hyperparameter $\bar{s}$ was set to 20. The FDR level was set to $\alpha_1 = 0.15$, which is relatively large such that the variable selection is liberal and thus reduces the risk of losing true predictors. In stage 2, the SAMC algorithm was run for the aggregated data set with $270p_n$ iterations, where the first $20p_n$ iterations were for the burn-in process. The gain factor sequence was set as $a_k = 10p_n/\max\{10p_n, k\}$, where k indexes the number of iterations of the SAMC algorithm. The prior hyperparameter $\bar{r}_n$ was set to 35. The FDR level was set to $\alpha_2 = 0.01$, which is small such that the resulting model is sparse.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

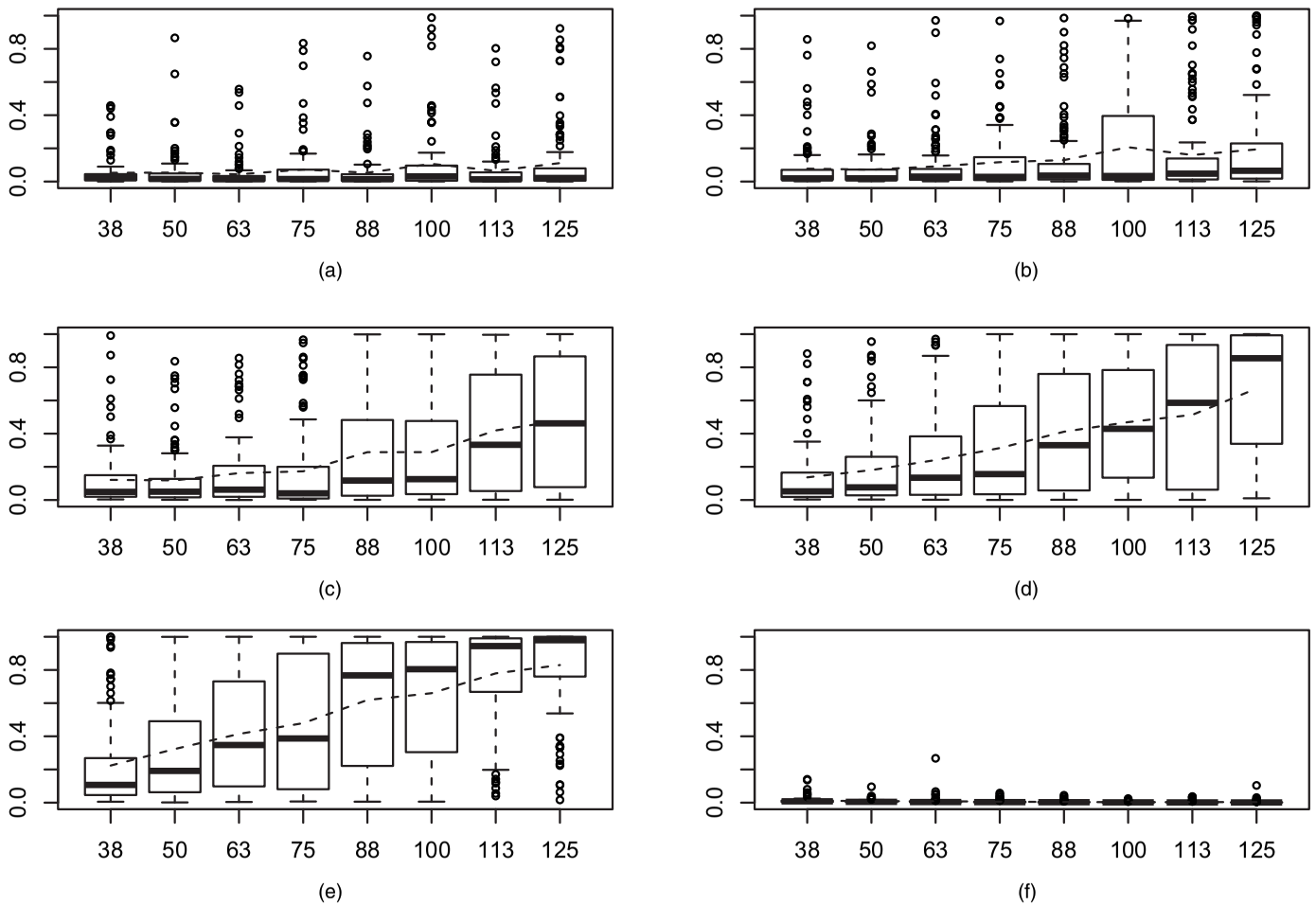


Fig. 3. Marginal inclusion probabilities of six predictors for the case of example 1 that the mutual correlation coefficient between different predictors is 0.9 for true regression coefficients (a) 0.7, (b) 0.9, (c) 1.1, (d) 1.3, (e) 1.5 and (f) 0: in each plot, the boxplots are for the sample sizes $n = 38, 50, 63, 75, 88, 100, 113, 125$; each boxplot shows the distribution of the marginal inclusion probabilities of one predictor calculated from 100 simulated data sets; ------, mean value of the marginal inclusion probabilities

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

Table 1. Results of the SAM approach for the second toy example†

Results for stage 1		*Results for stage 2*	
Size‡	*Number of correlated predictors*§	*Size*§§	*Number of true predictors**
166.6 (0.87)	98.2 (0.22)	3.2 (0.05)	3.0 (0)

†The values reported in parentheses are standard deviations of the corresponding estimate.
‡Average number of predictors selected in stage 1.
§Average number of predictors that are correlated with the true predictors and selected in stage 1.
§§Average number of predictors selected in stage 2.
*Average number of true predictors selected in stage 2.

Table 1 summarizes the simulation results. It shows that almost all predictors that are correlated with the true predictors are selected in stage 1, and all true predictors are selected in stage 2. The SAM approach works perfectly for this example.

4.2. Massive data example

This example compares the SAM approach with several popular penalized likelihood approaches for massive data sets. We simulated 100 data sets with $\sigma^{*2} = 0.25$. Each data set was generated from a multivariate normal distribution with $n = 100$ observations and $P_n = 500\,000$ predictors, among which 2000 predictors are equally correlated with a correlation coefficient of $\rho = 0.25$ and the rest of the predictors are uncorrelated. Among those 2000 correlated predictors, five were chosen as the true predictors with regression coefficients (0.2, 0.3, 0.4, 0.5, 0.6).

Each data set was randomly split into 1000 subsets with $s = 500$. The SAMC algorithm was then run for each subset of data and the aggregated subset data with the same setting as for example 2 of Section 4.1.1. In the simulations, we set $\bar{s} = 45$ as the upper bound of model size for the subset data and $\bar{r}_n = 35$ for the aggregated subset data, and we set the FDR level $\alpha_1 = 0.15$ for variable selection in stage 1 and $\alpha_2 = 0.01$ in stage 2. On average (over the 100 data sets), the SAM approach reduced the dimension of the data to 18057.7 in stage 1 and then selected 5.3 predictors in stage 2.

For comparison, several popular penalized likelihood approaches, including the lasso, elastic net, SIS and ISIS, were also applied to this example. The lasso was implemented for this example by using the R package `glmnet` (Friedman *et al.*, 2010) with the tuning parameter λ determined through a tenfold cross-validation procedure. The elastic net was also implemented by using the R package `glmnet`, for which we set the penalty function $p_\lambda = \lambda(\|\boldsymbol{\beta}\|_{L_1} + \|\boldsymbol{\beta}\|_{L_2}^2)$ and determined the value of λ via a tenfold cross-validation procedure. We used the R package `SIS` to implement SIS–SCAD and ISIS–SCAD, which first reduce the number of predictors to $n/\log(n)$ by marginal utility, and then apply SCAD to refine the model. The results are summarized in Table 2.

To measure the performance of different methods for this example, we calculated the false selection rate fsr and negative selection rate nsr. Let $\mathbf{t}$ denote the set of true predictors, and let ξ denote the model selected by a method. Then, fsr and nsr are defined by

$$\text{fsr} = |\xi \setminus \mathbf{t}|/|\xi|, \qquad \text{nsr} = |\mathbf{t} \setminus \xi|/|\mathbf{t}|,$$

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

Table 2. Comparison of the SAM approach with the lasso, elastic net, SIS and ISIS for the massive data example†

Parameter	*Results for the following methods:*				
	SAM	*Lasso*	*Elastic net*	*SIS*	*ISIS*
$\overline{\lvert\xi\rvert}$‡	5.3 (0.22)	44.62 (0.98)	52.70 (0.21)	11.47 (0.29)	13.57 (0.59)
$\overline{\lvert\xi \cap \mathbf{t}\rvert}$§	3.53 (0.073)	4.24 (0.070)	4.07 (0.074)	3.31 (0.083)	3.37 (0.086)
fsr (%)	25.5 (2.30)	89.7 (0.38)	92.3 (0.15)	67.7 (1.58)	65.8 (2.50)
nsr (%)	29.4 (1.46)	15.2 (1.40)	18.6 (1.48)	33.8 (1.65)	32.6 (1.72)
MSE§§	0.146 (0.013)*	0.263 (0.011)	0.488 (0.011)	0.246 (0.016)	0.342 (0.022)

†The numbers in parentheses are the corresponding standard deviations.
‡Average number of predictors selected by the different methods.
§Average number of true predictors selected by the different methods.
§§Mean-squared error of $\hat{\boldsymbol{\beta}}_\xi$, i.e. $\|\hat{\boldsymbol{\beta}}_\xi - \boldsymbol{\beta}^*\|^2$, produced by the different methods.
*$\boldsymbol{\beta}$ is estimated by Bayesian model averaging.

where $|\cdot|$ denotes the size of a model. The smaller the values of fsr and nsr are, the better the performance of the approach is. From Table 2, it is easy to see that the SAM method yields a much smaller value of fsr than all other methods, and about the same value of nsr as SIS and ISIS. The lasso and elastic net yield a smaller nsr but at the expense of a higher fsr. We have also compared the mean-squared errors of the estimates of $\boldsymbol{\beta}$ that are produced by different methods. For the SAM, approach $\boldsymbol{\beta}$ is estimated by averaging over the MCMC samples drawn in stage 2, while restricting the coefficients of the predictors that are screened out in stage 1 to 0. Strictly speaking, this estimator is not sparse, but, as shown in Table 2, it is more accurate than those produced by other methods.

A final remark for this massive data example is on its CPU cost. Under the same settings for the burn-in period and total iteration numbers as those used in Section 4.1, the SAM cost for each data set about 16 h in a serial implementation on a single core of Intel® Xeon® CPU E5-2690 (2.90 GHz). This converts to about half an hour if 100 such cores are run in parallel, after accounting for the CPU times that are spent on each subset and the aggregated subset.

5. Real data examples

5.1. Metabolic quantitative trait loci example

The first example is related to a metabolic quantitative trait loci experiment, which links SNPs data to metabolomics data (Dumas *et al.*, 2007). The predictors come from a genomewide analysis of candidate genes for alanine amino transferase enzyme elevation in liver with the mass spectroscopy metabolomics data as the response (Kindmark *et al.*, 2008). The spectra are divided by regions or bins to reduce the dimension of spectral data, and a common log-transformation is applied to normalize the signal. A total of 10000 SNPs are preselected as candidate predictors on the basis of the following criteria: no missing values, no monomorphic SNPs and close to known regulatory regions. The total number of subjects included in the data set is 50. The genotype of each SNP is coded as 0, 1 and 2 for homozygous rare, heterozygous and homozygous common allele respectively. As in Bottolo and Richardson (2010), one particular metabolite bin that discriminates between the disease status of the clinical trial's participants is chosen as the response.

The SAM approach was first applied to this example. The data set was divided into 20 subsets

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

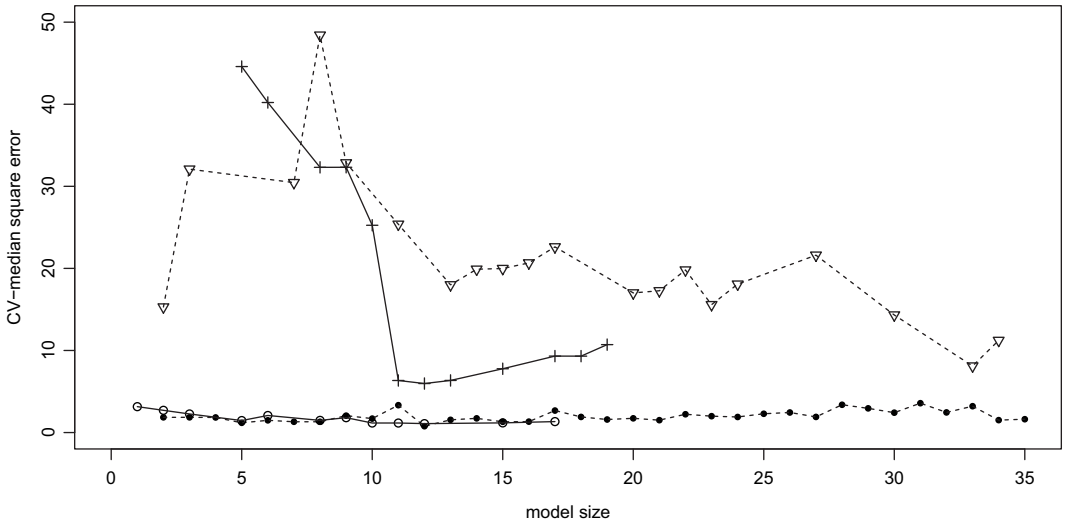


Fig. 4. Comparison of the SAM approach (•) with the lasso (∇), SIS–lasso (+) and the ISIS–lasso (○) for the metabolic quantitative trait loci data: leave-one-out cross-validation median-square error of the MAP_1–MAP_{35} models

with $s = 500$ and the method was then run with exactly the same setting as for the massive data example of Section 4.2. In stage 1 the SAM approach reduced the number of SNPs to 536 from 10 000, and in stage 2 it selected two SNPs, rs17041311 and rs17392161, whose marginal inclusion probabilities were 0.98 and 0.93 respectively. The two SNPs also compose the MAP model for this example. We note that, in the data set, the SNP rs7896824 has the same genotypes as the SNP rs17041311 across all 50 subjects. Similarly, the SNPs rs17390419, rs12328732, rs2164473, rs322664, rs17415876, rs16950829, rs6607364, rs829156, rs829157 and rs2946537 share the same genotypes with the SNP rs17392161.

For comparison, SIS and ISIS were applied to this example. SIS–SCAD and ISIS–SCAD first reduced the number of SNPs from 10 000 to 25 and then applied SCAD to refine the selection, but both yielded the null model. To assess the performance of different methods, we used leave-one-out cross-validation. The median cross-validation square error of the SAM model is 1.8, and that of the null model is 9.74. (There is a slight abuse of the term 'cross-validation' here. In this work, we use leave-one-out cross-validation to evaluate the predictive performance of a fixed model which is selected by some method. The mean cross-validation error of a fixed model is given by

$$\mathrm{CV}_\xi = \frac{1}{n}\sum_{i=1}^{n}(y_i - x_i^{\mathrm{T}}\hat{\beta}_\xi^{(i)})^2,$$

where ξ denotes the fixed model, $\hat{\beta}_\xi^{(i)}$ is the ordinary least squares estimator of β_ξ based on the data set $\{\mathbf{y}^{(-i)}, \mathbf{X}_\xi^{(-i)}\}$ and $(\cdot)^{(-i)}$ denotes removing the ith observation. The median cross-validation error is defined in a similar way.)

We also compared the MAP_i models that were produced by the SAM approach with those produced by the lasso, SIS–lasso and ISIS–lasso along their regularization paths. Here the MAP_i model refers to the MAP model containing i SNPs. Fig. 4 shows the leave-one-out median-square error of these models. The lasso and SIS–lasso failed to select the SNP rs17041311, which is one of the two important SNPs that were identified by the SAM approach, and thus yielded

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

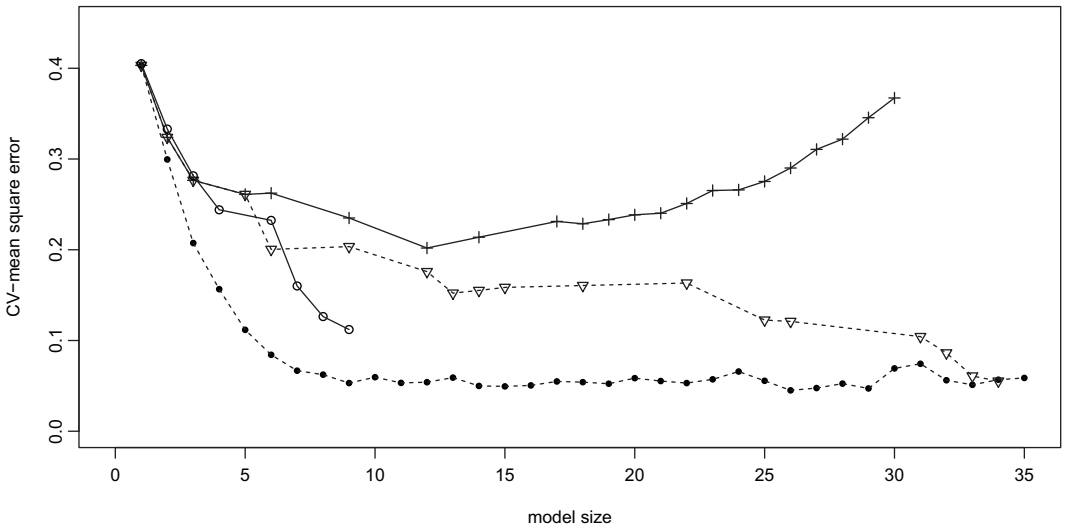


Fig. 5. Comparison of the SAM approach (•) with the lasso (∇), SIS–lasso (+) and the ISIS–lasso (○) for the polymerase chain reaction data: leave-one-out cross-validation median-square error of the MAP_1–MAP_{35} models

enormous median-square errors. The ISIS–lasso method successfully selected both rs17041311 and rs17392161, and thus yielded similar median-square errors to those of the SAM approach.

5.2. Polymerase chain reaction example

The second example relates to a polymerase chain reaction data set. Lan *et al*. (2006) conducted an experiment which examines the genetics of two inbred mouse populations (C57BL/6J and BTBR). A total of 60 F2 samples, with 31 female and 29 male mice, were used to monitor the expression levels of 22 575 genes. Some physiological phenotypes, including numbers of phosphoenopyruvate carboxykinase, glycerol-3-phosphate acyltransferase and stearoyl-CoA desaturase 1 were measured by quantitative realtime polymerase chain reaction. In this example, we study the relationship between phosphoenopyruvate carboxykinase (as responses) and the gene expression level. The gene expression data and the phenotype data are published in the gene expression omnibus database (`http://www.ncbi.nlm.nih.gov/geo`; accession number GSE3330). The gene expression data have been normalized before statistical analysis.

The SAM approach was first applied to this example. The data were divided into 45 subsets with $s = 502$. In stage 1, the method selected 1113 genes from 22 575 genes. In stage 2, the SAM approach selected six genes under the setting $r_n = P_n^{0.3}$. The six selected genes are 1429089_s_at, 1430779_at, 1432745_at, 1437871_at, 1440699_at and 1459563_x_at. The first five genes also compose the MAP model. The leave-one-out cross-validation mean-square error of the model of the six genes is 0.084. In comparison, SIS–SCAD selected 17 genes with leave-one-out mean-square error 0.204, and ISIS–SCAD selected nine genes with leave-one-out mean-square error 0.112.

Fig. 5 shows the leave-one-out mean-square errors of the MAP_i models selected by the SAM approach and the models selected by the lasso, SIS–lasso and ISIS–lasso along their regularization paths. For this example, the SIS–lasso and ISIS–lasso first reduced the number of genes to 30 and then applied the lasso to refine the selection. From Fig. 5, it is easy to see that the SAM approach significantly outperforms the penalized likelihood approaches for this example.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

6. Time complexity analysis

Since the SAM approach has an embarrassingly parallel structure, its time complexity can be simply analysed as follows. Suppose that there are enough numbers of processors available in the parallel architecture, and each subset is of about the same size. Let K_n denote the number of subsets, i.e. the number of processors that are used in SAM implementation. Then each subset contains about P_n/K_n predictors. Let $\bar{s}_n$ denote the upper bound of the size of the models that are considered for each subset. Then, for each subset, the computational complexity of the first-stage screening is bounded by

$$O(nP_n^2/K_n^2) + \{O(\bar{s}_n^3) + O(P_n/K_n)\}T,$$

where the first term is for the overhead computation of the matrix or vector products $\mathbf{X}_\mathbf{s}^\mathrm{T}\mathbf{X}_\mathbf{s}$, $\mathbf{X}_\mathbf{s}^\mathrm{T}\mathbf{y}$ and $\mathbf{y}^\mathrm{T}\mathbf{y}$, which are prepared for calculation of $(\mathbf{X}_\xi^\mathrm{T}\mathbf{X}_\xi)^{-1}$ and $\mathbf{X}_\xi^\mathrm{T}\mathbf{y}$ in simulations of the posterior (7), T is the number of iterations, $O(\bar{s}_n^3)$ is for evaluation of the posterior (7), which involves inversion of the matrix $\mathbf{X}_\xi^\mathrm{T}\mathbf{X}_\xi + V_\xi^{-1}$, and $O(P_n/K_n)$ is for evaluation of the proposal distribution as described in the on-line supplementary material. In our implementation of the SAMC algorithm we employed a proposal distribution for which the computational time for adding to or exchanging with the current model a predictor is approximately proportional to the number of predictors contained in the subset. If we further let the number of iterations increase with the subset size in a low power, say $T = O\{(P_n/K_n)^\alpha\}$ for some $\alpha \geqslant 1$, then the computational complexity for each subset is $O\{(P_n/K_n)^{1+\alpha}\} + O\{n(P_n/K_n)^2\}$ by assuming that $\bar{s}_n = o\{(P_n/K_n)^{1/3}\}$ holds. To maintain the same estimation accuracy of marginal inclusion probabilities, α should be no less than 1. In this paper, we set $\alpha = 1$. Similarly, we can derive that the computational complexity for the aggregated subset is $O\{(K_n\bar{r})^{1+\alpha}\} + O\{n(K_n\bar{r})^2\}$, where $\bar{r}$ denotes the average number of predictors selected from each subset, and $\bar{s}$ denotes the upper bound of the final model size.

If we choose K_n to minimize the time complexity of the SAM approach, i.e. minimizing

$$O\{(P_n/K_n)^{1+\alpha}\} + O\{(K_n\bar{r})^{1+\alpha}\} + O\{n(P_n/K_n)^2\} + O\{n(K_n\bar{r})^2\},$$

then it is easy to verify that the time complexity can be minimized at

$$K_n = O(\sqrt{P_n}),$$

for any $\alpha > 0$. If $K_n = O(\sqrt{P_n})$, then the resulting subset size is $O(\sqrt{P_n})$, and the resulting time complexity of the SAM approach is $O(P_n^{(1+\alpha)/2}) + O(nP_n)$, i.e. the SAM approach can work in a linear or low order polynomial time of P_n. This is rather attractive compared with the existing high dimensional variable selection approaches. For example, the lasso has a computational complexity of $O(nP_n^2)$ when $P_n > n$ (Meinshausen, 2007). In a parallel implementation (see for example Boyd *et al.* (2010)) with $\sqrt{P_n}$ processors, its time complexity should be lower bounded by $O(nP_n^{3/2})$. The elementwise Gibbs sampler with a local shrinkage prior has a time complexity of at least $O(P_n^4)$, given that its computational complexity at each iteration is $O(P_n^3)$. Even in a parallel implementation with $\sqrt{P_n}$ processors, its time complexity should be lower bounded by $O(P_n^{7/2})$.

It is interesting to point out that, even if implementing on a serial computer, the SAM approach can still have a favourable time complexity compared with the direct simulation of the full data posterior (i.e. setting $K_n = 1$). The time complexity of the SAM approach in a serial implementation is $\sqrt{P_n}\, O(P_n^{(1+\alpha)/2} + nP_n)$ if $\sqrt{P_n}$ subsets are used, whereas the time complexity of the direst simulation of the full data posterior is $O(P_n^{1+\alpha}) + O(nP_n^2)$. The SAM strategy leads to a CPU time saving of order at least $O(\sqrt{P_n})$ in serial implementations!

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

In the recent literature, some researchers have explored the use of graphics processing units for scaling up Bayesian computation; see for example Suchard *et al*. (2010), Lee *et al*. (2010) and Bottolo *et al*. (2013). It is obvious that the SAM method can benefit from the use of graphics processing units if parallel architectures are not available. For example, following Lee *et al*. (2010), one can implement a population SAMC algorithm (Song *et al*., 2014) with graphics processing units for each subset of data. Following Bottolo *et al*. (2013), one can use graphics processing units to accelerate the computation of the likelihood function in SAMC simulations.

7. Discussion

In this paper, we propose a new Bayesian variable selection approach, the so-called SAM approach, for ultrahigh dimensional linear regression. The SAM approach works in two stages: stage 1 screens out the predictors that are uncorrelated with the response variable, and stage 2 refines the selection of predictors. Both stages work under a Bayesian framework. Compared with the SIS approach of Fan and Lv (2008), a significant advantage of the SAM approach is that it makes use of the joint information of multiple predictors in predictor screening, whereas SIS makes use of only the marginal utility of each predictor. Our numerical results, on both simulated and real data sets, show that the SAM approach can significantly outperform SIS and ISIS, and also significantly outperforms other penalized likelihood approaches such as the lasso and elastic net. The models that are selected by the SAM approach are more sparse and closer to the true model.

The SAM approach has an embarrassingly parallel structure and can be easily implemented in a parallel architecture. Therefore, it is ready to be applied to big data problems with millions or more predictors. This has been beyond the ability of conventional Bayesian approaches that directly work on the full data set. At this point, the SAM approach also distinguishes from SIS. Although SIS does penalized regression on a reduced data set, it still needs to process all predictors altogether in the screening stage—sorting the marginal utilities of all predictors. However, the SAM approach never needs to process all the predictors altogether. From this point of view, it is more suitable for big data problems.

We note that SAM is not a standard Bayesian approach, as it does not process all data altogether and thus only provides an approximate quantification of uncertainty for each predictor through marginal inclusion probabilities. Since the marginal inclusion probability measures the contribution of a predictor given all other predictors in the regression, the marginal inclusion probability of a given predictor in the subset data regression should not be less than that in the full data regression, although the full data regression is never run in the SAM method. This observation is meaningful particularly for the predictors that are screened out in the first stage, for which the marginal inclusion probabilities that are obtained in the subset data regression provide an upper bound for their marginal inclusion probabilities in the full data regression.

Regarding how to interpret the posterior probabilities that are associated with Bayesian model averaging in the subset data regression in comparison with their counterparts in the full data regression, we note that their difference can be interpreted as a difference resulting from their priors on the model space. The subset data regression is equivalent to the use of a data-dependent prior, which sets the prior probability to 0 for any models containing predictors that have been screened out in stage 1. Alternatively, the estimator that is obtained in the subset data regression can be interpreted as an Occam's razor approximation (Madigan and Raftery, 1994) to its counterpart in the full data regression, as the models containing the predictors that are screened out in stage 1 tend to have very low posterior probabilities.

Although the SAM approach does not work in a standard Bayesian way, it controls the model

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

sparsity as in other Bayesian methods through specifying model priors, i.e. specifying *a priori* selection probability λ_n. The multiplicity is controlled by setting $\lambda_n \to 0$ as $P_n \to \infty$. Hence, as implied by expression (6), the prior of the SAM approach works more like an L_0-penalty. This may be a partial reason why the SAM approach produces favourable results compared with the lasso and SIS. The latter two methods employ an L_1- or SCAD penalty which both impose nearly 0 penalty values on nearly 0 coefficients, and thus the resulting models tend to be dense.

Regarding parallel implementations for high dimensional variable selection, we note that Boyd *et al*. (2010) proposed a parallel algorithm for the lasso and group lasso by splitting across features. Although both Boyd's algorithm and the SAM approach work on the basis of the same strategy of feature splitting, they are fundamentally different. The SAM approach is more on the side of statistics; it is a new statistical algorithm which has an embarrassingly parallel structure, and new statistical theory is required for its performance. Although Boyd's algorithm is mainly on the side of computer science, it just implements the existing methods in a parallel way. Since Boyd's algorithm cannot be implemented in an embarrassingly parallel way, internode communications may significantly reduce its efficiency. In addition, as discussed in Section 6, Boyd's algorithm may have a higher time complexity than the SAM method.

To justify the SAM approach, we establish Bayesian posterior consistency under both situations that the model is correctly specified and misspecified. We show that, when the model is correctly specified, the marginal inclusion probability of the true predictor will converge to 1 as the sample size becomes large; and, when the model is misspecified, the marginal inclusion probability of the predictor that is correlated with the response will converge to 1 as the sample size becomes large. We also show the sure screening property for the predictors that are selected on the basis of the marginal inclusion probability and that the MAP model is consistent as an estimator of the true model.

In this paper, the theoretical results are established under a Gaussian prior setting for the regression coefficients. Although this prior setting has been widely used in the literature (e.g. Jiang (2005) and Liang *et al*. (2013)), it is not natural from the perspective of marginal inclusion probability evaluation as it assigns the highest density value to the null point. It is of interest to use a non-local prior (Johnson and Rossel, 2012), which assigns a zero probability to the null point and may lead to a better rate of convergence of the posterior distribution. A further extension of the SAM approach to generalized linear models is also of great interest.

Finally, we note that, in the recent literature, there is an active development of Bayesian sparse learning for high dimensional variable selection with local shrinkage priors (see for example Polson and Scott (2012)), such as the Laplace prior (Park and Casella, 2008) and the horseshoe prior (Carvalho *et al*., 2009, 2010), which aim to produce a sparse model by analogy with the pure shrinkage approach of something like the lasso under the Bayesian framework. We have conducted a comparison study to compare the performance of the horseshoe prior and the prior that is used in this paper. The latter is also called the SAM prior in the on-line supplementary material. As shown in Fig. 1 of the supplementary material, the SAM prior can perform much better than the horseshoe prior. The SAM prior is posterior consistent and efficient in computation by restricting the models to be of a manageable size, whereas the horseshoe prior involves sampling from high dimensional normal distributions, and it is unclear whether it is posterior consistent or not under the high dimensional setting.

8. Supplementary material

The supplementary material is organized as follows. Section 1 gives the proofs of theorems 1–4, section 2 describes the SAMC algorithm and its implementation for high dimensional re-

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

gression, section 3 establishes the consistency of the MAP model under the posterior distribution (7) and section 4 presents a comparison study for the performance of the horseshoe prior and the SAM prior.

Acknowledgements

The authors thank Dr Leonardo Bottolo for providing the metabolic quantitative trait loci data set and thank the Joint Editor, Associate Editor and two referees for their constructive comments which have led to significant improvement of this paper. Liang's research was partially supported by National Science Foundation grants DMS-1106494 and DMS-1317131.

References

Barbieri, M. and Berger, J. (2004) Optimal predictive model selection. *Ann. Statist.*, **32**, 870–897.
Berk, R. H. (1966) Limiting behaviour of posterior distributions when the model is incorrect. *Ann. Math. Statist.*, **37**, 51–58.
Bickel, P. and Levina, E. (2004) Some thoery for Fisher's linear discriminant function, 'naive Bayes', and some alternatives when there are many more variable than observations. *Bernoulli*, **10**, 989–1010.
Bondell, H. D. and Reich, B. J. (2012) Consistent high-dimensional Bayesian variable selection via penalized credible regions. *J. Am. Statist. Ass.*, **107**, 1610–1624.
Bottolo, L., Chadeau-Hyam, M., Hastie, D. I., Zeller, T., Liquet, B., Newcombe, P., Yengo, L., Wild, P. S., Schillert, A., Ziegler, A., Nielsen, S. F., Butterworth, A. S., Ho, W. K., Castagné, R., Munzel, T., Tregouet, D., Falchi, M., Cambien, F., Nordestgaard, B. G., Fumeron, F., Tybjdaerg-Hansen, A., Froguel, P., Danesh, J., Petretto, E., Blankenberg, S., Tiret, L. and Richardson, S (2013) GUESS-ing polygenic associations with multiple phenotypes using a GPU-based evolutionary stochastic search algorithm. *PLOS Genet.*, **9**, article e1003657.
Bottolo, L. and Richardson, S. (2010) Evolutionary stochastic search for Bayesian model exploration. *Baysn Anal.*, **5**, 583–618.
Boyd, S., Parikh, N., Chu, E., Peleato, B. and Eckstein, J. (2010) Distributed optimization and statistical learning via the alternating direction method of multipliers. *Found. Trends Mach. Learn.*, **3**, 1–122.
Broman, K. W. and Speed, T. P. (2002) A model selection approach for the identification of quantitative trait loci in experimental crosses. *J. R. Statist. Soc.* B, **64**, 641–656.
Carvalho, C., Polson, N. and Scott, J. (2009) Handling sparsity via the horseshoe. *J. Mach. Learn. Res.*, **5**, 73–80.
Carvalho, C., Polson, N. and Scott, J. (2010) The horseshoe estimator for sparse signals. *Biometrika*, **97**, 465–480.
Chen, J. and Chen, Z. (2008) Extended Bayesian information criteria for model selection with large model spaces. *Biometrika*, **95**, 759–771.
De Blasi, P. and Walker, S. (2013) Bayesian asymptotics with misspecified models. *Statist. Sin.*, **23**, 169–187.
Dumas, M., Wilder, S., Bihoreau, M., Barton, R., Fearnside, J., Arqoud, K., D'Amato, L., Wallis, R., Blancher, C., Keun, H., Baunsqaard, D., Scott, J., Sidelmann, U., Nicholson, J. and Gauquier, D. (2007) Direct quantitative trait locus mapping of mammalian metabolic phenotypes in diabetic and normoglycemic rat models. *Nat. Genet.*, **39**, 666–672.
Fan, J. and Li, R. (2001) Variable selection via nonconcave penalized likelihood and its oracle properties. *J. Am. Statist. Ass.*, **96**, 1348–1360.
Fan, J. and Lv, J. (2008) Sure independence screening for ultrahigh dimensional feature space (with discussion). *J. R. Statist. Soc.* B, **70**, 849–911.
Fan, J. and Song, R. (2010) Sure independence screening in generalized linear model with NP-dimensionality. *Ann. Statist.*, **38**, 3567–3604.
Friedman, J., Hastie, T. and Tibshirani, R. (2010) Regularization paths for generalized linear models via coordinate descent. *J. Statist. Softwr.*, **33**, 1–22.
Guan, Y. and Stephens, M. (2011) Bayesian variable selection regression for genome-wide association studies and other large-scale problems. *Ann. Appl. Statist.*, **5**, 1780–1815.
Gupta, M. and Ibrahim, J. (2009) An information matrix prior for Bayesian analysis in generalized linear models with high dimensional data. *Statist. Sin.*, **19**, 1641–1663.
Haario, H., Saksman, E. and Tamminen, J. (2001) An adaptive Metropolis algorithm. *Bernoulli*, **7**, 223–242.
Jiang, W. (2005) Bayesian variable selection for high dimensional generalized linear models. *Technical Report 05-02*. Department of Statistics, Northwestern University, Evanston.
Jiang, W. (2007) Bayesian variable selection for high dimensional generalized linear models: convergence rate of the fitted densities. *Ann. Statist.*, **35**, 1487–1511.
Johnson, V. (2012) On numerical aspects of Bayesian model selection in high and ultrahigh-dimensional settings. *Baysn Anal.*, **7**, no. 4, 1–18.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026

Johnson, V. and Rossel, D. (2012) Bayesian model selection in high-dimensional settings. *J. Am. Statist. Ass.*, **107**, 649–660.
Kindmark, A., Jawaid, A., Harbron, C., Barrat, B. and March, R. (2008) Genome-wide pharmacogenetic investigation of a hepatic adverse event without clinical signs of immunopathology suggests an underlying immune pathogenesis. *Pharmgenom. J.*, **8**, 186–195.
Kleijn, B. and van der Vaart, A. (2006) Misspecification in infinite-dimensional Bayesian statistics. *Ann. Statist.*, **34**, 837–877.
Kundu, S. and Dunson, D. B. (2014) Bayes variable selection in semiparametric linear models. *J. Am. Statist. Ass.*, **109**, 437–447.
Lan, H., Chen, M., Flowers, J., Yandell, B., Stapleton, D., Mata, C., Mui, E.., Flowers, M., Schueler, K., Manly, K., Williams, R., Kendziorski, K. and Attie, A. D. (2006) Combined expression trait correlations and expression quantitative trait locus mapping. *PLOS Genet.*, **2**, article e6.
Lee, A., Yau, C., Giles, M., Doucet, A. and Holmes, C. (2010) On the utility of graphics cards to perform massively parallel simulation of advanced Monte Carlo methods. *J. Computnl Graph. Statist.*, **19**, 769–789.
Li, F. and Zhang, N. (2010) Bayesian variable selection in structured High-Dimensional covariate spaces with application in genomics. *J. Am. Statist. Ass.*, **105**, 1202–1214.
Liang, F. (2009) On the use of SAMC for Monte Carlo intergration. *Statist. Probab. Lett.*, **79**, 581–587.
Liang, F., Liu, C. and Carroll, R. (2007) Stochastic approximation in Monte Carlo computation. *J. Am. Statist. Ass.*, **102**, 305–320.
Liang, F., Paulo, R., Molina, G., Clyde, M. and Berger, J. (2008) Mixture of g priors for Bayesian variable selection. *J. Am. Statist. Ass.*, **103**, 410–423.
Liang, F., Song, Q. and Yu, K. (2013) Bayesian subset modeling for high dimensional generalized linear models. *J. Am. Statist. Ass.*, **108**, 589–606.
Liang, F. and Zhang, J. (2008) Estimating FDR under general dependence using stochastic approximation. *Biometrika*, **95**, 961–977.
Lin, N. and Xi, R. (2011) Aggregated estimation equation estimation. *Statist. Interf.*, **4**, 73–83.
Mackey, L., Talwalkar, A. and Jordan, M. (2012) Divide-and-conquer matrix factorization. *Preprint arXiv: 1107.0789*.
Madigan, D. and Raftery, A. (1994) Model selection and accounting for model uncertainty in graphical models using Occam's windows. *J. Am. Statist. Ass.*, **89**, 1535–1546.
Meinshausen, N. (2007) Relaxed Lasso. *Computnl Statist. Data Anal.*, **52**, 374–393.
Park, T. and Casella, G. (2008) The Bayesian Lasso. *J. Am. Statist. Ass.*, **103**, 681–686.
Polson, N. G. and Scott, J. G. (2012) Local shrinkage rules, Lévy processes and regularized regression. *J. R. Statist. Soc.* B, **74**, 287–311.
Scott, J. and Berger, J. (2010) Bayes and empirical-Bayes multiplicity adjustment in the variable-selection problem. *Ann. Statist.*, **38**, 2587–2619.
Shalizi, C. (2009) Dynamics of Bayesian updating with dependent data and misspecified models. *Electron. J. Statist.*, **3**, 1039–1074.
Song, Q., Wu, M. and Liang, F. (2014) Weak convergence rates of population versus single-chain stochastic approximation MCMC algorithms. *Adv. Appl. Probab.*, to be published.
Strawn, N., Armagan, A., Saab, R., Carin, L. and Dunson, D. (2012) Finite sample posterior concentration in high-dimensional regression. *Preprint arXiv:1207.4854*.
Suchard, M., Wang, Q., Chan, C., Frelinger, J., Cron, A. and West, M. (2010) Understanding GPU programming for statistical computation: studies in massively parallel massive mixtures. *J. Computnl Graph. Statist.*, **19**, 419–438.
Tibshirani, R. (1996) Regression shrinkage and selection via the lasso. *J. R. Statist. Soc.* B, **58**, 267–288.
Wilson, M. A. (2010) Bayesian model uncertainty and prior choice with applications to genetic association studies. *PhD Thesis*. Duke University, Durham.
Xi, R., Lin, N. and Chen, Y. (2009) Compression and aggregation for logistic regression analysis in data cubes. *IEEE Trans. Knowl. Data Engng*, **21**, 479–492.
Zellner, A. (1986), On assessing prior distributions and Bayesian regression analysis with g-prior distribution. In *Bayesian Inference and Decision Techniques: Essays in Honor of Bruno de Finetti* (eds P. Goel and A. Zellner), pp. 233–243. Amsterdam: North-Holland.
Zhang, C.-H. (2010) Nearly unbiased variable selection under minimax concave penalty. *Ann. Statist.*, **38**, 894–942.
Zou, H. (2006) The adaptive lasso and its oracle properties. *J. Am. Statist. Ass.*, **101**, 1418–1429.
Zou, H. and Hastie, T. (2005) Regularization and variable selection via the elastic net. *J. R. Statist. Soc.* B, **67**, 301–320.

Supporting information

Additional 'supporting information' may be found in the on-line version of this article:

'Supplementary materials for "A split-and-merge variable selection approach for ultra-high dimensional regression"'.

Downloaded from academic.oup.com/jrsssb/article/77/5/947/7040600 by Rice University user on 03 November 2026